

VOL. II.

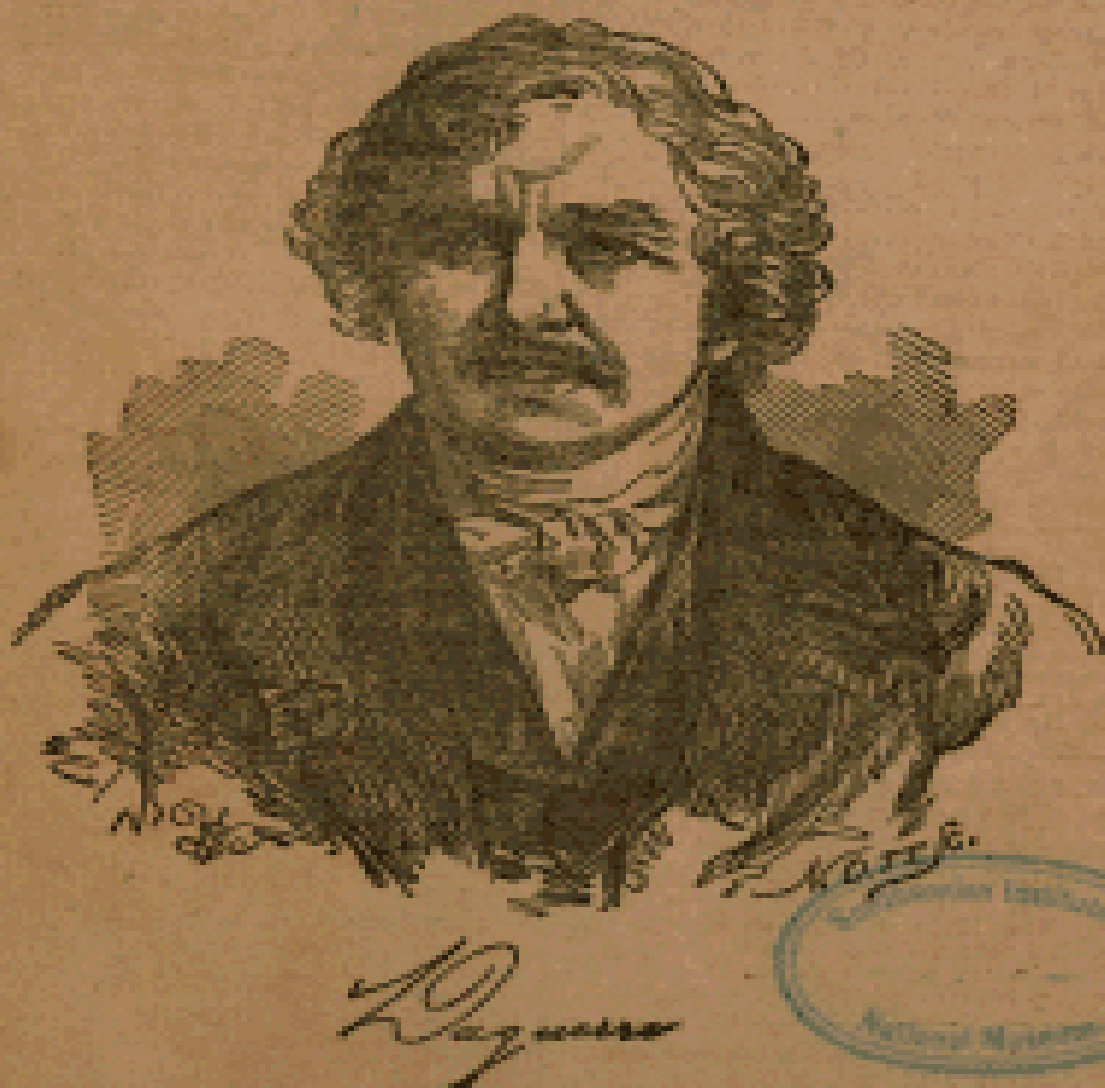
JUNE 1, 1851.

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No. 2.

THE
DAGUERREIAN JOURNAL:

Devoted to the Daguerreian and Photogenic Arts.

Also, embracing the Sciences, Arts, and Literature.



S. D. HUMPHREY AND L. L. HILL, EDITORS.

NEW YORK:

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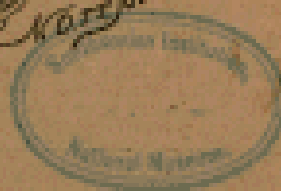
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Yours respectfully,
Albert Litch,
Formerly of the firm of LITCH & WHIPPLE, Boston, Mass.

THE DAGUERREIAN JOURNAL.

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NEW YORK, JUNE 1, 1851.

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SOME EXPERIMENTS AND REMARKS
ON THE CHANGES WHICH BODIES ARE CAPABLE OF
UNDERGOING IN DARKNESS, AND ON THE AGENT
PRODUCING THESE CHANGES.

BY ROBERT HUNT.

[Concluded.]

With the view of testing Dr. Draper's results, I carefully iodized two silver plates and exposed them to light. I then placed them so that half of one plate was covered by half of the other, and allowed them to remain in the dark 1-24th of an inch apart for four hours. On mercurialization I could not detect the slightest difference between the covered and uncovered portions of either of the plates.

Another silver plate was iodized and exposed to light. It was then placed in the dark with a sensitive plate which had been carefully kept from the light, 1-16th of an inch above it, and a small engraving placed between them. They were allowed to remain thus for six hours. When exposed to the vapor of mercury, the plate which had been subjected to the light whitened all over, and the space occupied by the engraving was distinctly marked by lines of vapor thicker than the other parts. The plate which had been preserved in the dark was scarcely at all influenced by the vapor, except on those parts which had been touched by the supports of card-board on which it rested. These were so arranged that no radiation could have influenced those parts of the plates.

An iodized silver plate was placed in the dark with a little fine string coiled over parts of it, and a polished silver plate supported 1-8th of an inch above it. After four hours both plates were subjected to mercurial vapor. On the iodized plate the deposit of vapor was uniform, although slight; but on the superposed plate of silver a strong and beautiful image of the string on the under plate became visible. I found that neither of the two iodized plates had lost their sensitiveness by the operations to which they had been subjected in the dark.

Hoping to detect some evidence of the process by which these singular results were produced, I instituted a series of experiments, of which the following are some

of the most interesting results.

A. A silver plate was iodized, a piece of card was placed upon it, and a well-polished mercurial plate (amalgamated copper) was suspended 1-8th of an inch above it, and left in this state for a night. The space on the silver plate corresponding with the mercurial plate, except under the card, was nearly freed of its iodine, which had evidently combined with the mercury on the upper plate. On exposing the mercurial plate to the vapor of mercury the image of the card was rendered visible, the vapor covering every part of the plate except that opposite the card. The silver plate received the vapor only on those parts which were not influenced by the mercurial plate. The upper plate was suspended by strings; these were faithfully imaged on both plates; by a thick line of mercurial vapor on the under plate, by the absence of it in the upper one.

B. An iodized *silvered* plate was exposed to light until brown, and a mercurial plate suspended above it for twelve hours. The browned silver plate was *whitened*, and all the irregularities of the mercurial plate strikingly marked on it: the mercurial plate was slightly tarnished. On rubbing the silvered plate it was found that the silver was removed more readily over the whitened portion, but had lost none of its adhesion in other parts.

C. Over an iodized silver plate, plates of gold, platina, silver, brass, copper, copper amalgamated, and zinc were placed at the distance of 1-8th of an inch. After three hours the amalgamated plate had made a decidedly visible impression on the silver one. On exposure to vapor, the mercury lodged on every part of the plate except that affected by the mercurial plate; some irregularities were observed, but none which could be decidedly traced to the other metals in juxtaposition. I have some evidence that different metals near each other seriously interfere with each other's influence.

D. A mercurial plate was iodized, and another mercurial plate placed 1-8th of an inch above it. The upper plate became covered with a bright yellow film; and on exposing them to mercurial vapor, marks became apparent which corresponded with those in the opposite plate.

E. A silver plate was iodized and placed in the dark with an engraving, face down, upon it. An amalgamated copper plate was laid on this, and left for fifteen hours. The mercurial plate was reddened, and on exposure to the vapor of mercury, a very nice impression of the engraving was brought out, it having been effected through the thickness of the paper. On the silvered plate the space covered by the paper was well marked; but vaporization produced no trace of the engraving. The space beyond the paper was rendered white. It was curious that both plates had several spots which corresponded, particularly two, distinguished by a well-defined

circle and a comet-like appendage, in length ten times the diameter of the circle. These spots could not be traced to anything visible in the print or either of the plates, and must, I think, be referred to some electrical influence. I find it indeed commonly the case, that the plates, after being subjected to these kind of experiments a few times, become mottled, or present on their polished faces all the appearances of a finely-grained wood, and in this state they are less susceptible of receiving any impression than when not so.

F. A silver plate was iodized and placed upon an engraving laid on a brightly polished mercurial plate, and left in the dark for twenty-four hours. The mercurial plate was turned brown, and the silver plate was left in the same state as if it had been exposed to sunshine, being *brown and black*. Neither of these plates gave a copy of the picture.

G. A mercurial plate was iodized, and above it was placed a plate of polished iron, a disc of paper being first laid on the mercurial plate, and they were left in this state for some hours. On exposing the iron plate to mercurial vapor, it was abundantly lodged over that space opposite the paper disc, but not at all on the other parts. The mercurial plate was attacked by vapor over every part but that which the paper disc protected.

Lead and zinc plates were used instead of the iron one, with nearly similar results.

H. A Daguerreotype was taken, and without removing the iodine a mercurial plate was placed a little above it, and left for ten hours. When removed, well-defined traces of the Daguerreotype picture were evident on the mercurial plate, which leads me to hope that by careful manipulation we may succeed in multiplying these beautiful productions by an easy method.

I became desirous of ascertaining whether the mercurial plates would produce any change upon the precipitated iodide of silver. I find by many experiments, that if the iodide of silver is pure, no more change is produced than is produced upon it by diffused light; but if it is rendered sensitive by a trace of the nitrate of silver, it is then darkened as by solar influence.

Sensitive iodide of silver being placed upon a plate of glass, a mercurial plate was fixed 1-8th of an inch above it. In three days the iodide of silver had become a deep brown, almost a black, and the mercurial plate was covered with the yellow iodide of mercury. Nitric acid dissolved the dark portion of the silver salt, as did also ammonia, on which was formed Faraday's oxide of silver, thereby proving the change, either by a primary or a secondary process, of the iodide into the oxide of silver. This experiment has been repeated at least a dozen times, and always with the same results. If a little heap of the iodide of silver is placed under a mercurial plate, it

is exceedingly interesting to witness the gradual formation of the very beautiful colored rings on the mercury in the progress of its conversion into an iodide. By prolonged action the yellow iodide passes into the bright red biniodide of mercury. I have some experiments now in hand, which convince me that similar chemical changes are to be effected through considerable spaces. I have succeeded in decomposing the iodide of copper and the iodide of gold by mercurial plates placed nearly a quarter of an inch above them.

I have an extensive record of results similar to those I have now detailed, all of them showing that the changes brought about by this mysterious agent, whether it be heat, light, or an undiscovered element, cannot be referred to those rays which the admirable researches of Sir John Herschel have shown to be the operative ones in producing the photographic phenomena which have so interested the world by their novel beauty, and which professor Draper includes within his general term—tithonicity. With regard to the detithonizing influence of the gases mentioned by Dr. Draper in his paper in your March number, I can only consider the results, which I find to be as he has stated, as the simple reconversion of the decomposed iodide of silver into another definite chemical compound. An iodized plate is exposed to light, the iodide of silver or other sensitive salt is decomposed, and in a state to receive mercurial vapor. It is now passed through an atmosphere of iodine, of chlorine, of bromine, or of nitrous gas. Chemists are well aware of the surprising energy with which these bodies attack the metals, consequently the exposure of a moment is quite sufficient to convert the surface which has undergone a change, into an iodide, chloride, bromide, or nitrite of silver. I certainly cannot see the necessity of going so far out of our way for an explanation of this effect as Dr. Draper has done.

I fear I have already occupied too much of your valuable space, or I might be inclined to trespass further. I shall, however drop my pen for the present, again assuring you that I only desire to keep the image of Truth which is just shadowing our path, as free as possible from mists which might in any way obscure it.—*Philosophical Magazine*, 1843.



RAILROAD TO THE PACIFIC.—Forest Shepherd, of Sacramento City, says it is now ascertained almost beyond doubt, that a railroad can be constructed from the Mississippi to the Pacific, without crossing any mountains, or meeting more impediment from snow than between Albany and Boston.



EXPERIMENTS ON THE COLORED FILMS

FORMED BY IODINE, BROMINE, AND CHLORINE UPON VARIOUS METALS.

BY AUGUSTUS WALLER, M. D.

In a paper presented by me to the Academy of Sciences of Paris, an extract from which may be seen in the *Comptes Rendus* for October 5, 1840, I first demonstrated the error committed in ascribing to the iodide of silver alone the power of fixing the vapors of mercury, after it had been exposed to the action of light. Instead of this property being exclusively confined to a film of iodide of silver, as obtained in the process of M. Daguerre, I found that it existed in many other substances when presented to the action of light in the state of thin films, viz. by the bromide and chloride of silver; by the oxide, bromide, iodide and chloride of copper, and some others; all these, however, possessing less sensibility than the iodide of silver of Daguerre, and therefore less available for the reproduction of the images of the camera than the compound originally discovered by that gentleman. The iodide of Daguerre was found already too little sensitive to the influence of light in this climate, especially when applied to the reproduction of the image of animate objects, so that those films discovered by me seemed still less suitable to be employed for that purpose; this objection has, however, been completely removed by recent improvements, more particularly those of M. Claudet, who effected this principally by combining the original discovery of Daguerre with those mentioned above as having been subsequently made by myself. Pursuing the first stage of Daguerre's process, he obtained the film of iodide of silver, and added to this another film of bromide, either in a simple state,—as practised in my experiments published more than six months before—or after two of these substances had been combined together, as the chloride of iodine and the bromide of iodine, which he was the first to employ.

These colored films, however, merit attention independently of the purposes to which they may be applied in photography: the beauty of some of the phenomena themselves is peculiarly attractive; the numerous changes of color they undergo, either by a variation in the thickness of the film, or by the action of light, assign them a place among the most curious facts of science, and the extreme facility with which they are obtained adds to the interest they excite.

Impressed with these ideas, I was induced to pursue a train of investigation on this subject; among the results of which, one of the most interesting was a new method of making colored rings, like those generally known under the name of "Newton's colored rings," on many of the metals, by the same chemical process as that employed for forming the films of uniform thickness in photography. In order to procure these colored rings, and at the same time to show the identity of the origin of the colors with those of the ordinary transparent films, that is, as residing simply in the thickness of the lamina and not dependent on the ordinary cause of color, we have but to place a piece of iodine on a well-polished surface of silver or copper, and in a short time we find around the iodine a series of colored zones of the various tints of the spectrum, and approaching in a greater or less degree to the form of a circle, according as they have been more or less disturbed in their formation by currents of the surrounding air. In order that they may be perfectly regular, as large as possible, and with tints undisturbed by the action of light, it is necessary to place a piece of iodine in the centre of a well-polished plate, as before described; this is then to be shaded by an opaque screen superimposed a few lines from the surface to cause the vapors which would otherwise ascend and partially escape, to expand over its silver surface. Colored rings may be formed in the same manner by bromine and chlorine and the various combinations of these bodies with each other, except that for those that are gaseous or liquid it is requisite to pay a little attention to the manner of disengaging them on the surface of the metal, either by passing them through a glass tube, or by some other contrivance easy to execute.

These rings correspond to those formed by reflected light in Newton's experiments, with this difference, however, that in the colored films of the soap bubble, and in those formed by the glass lenses, the thinnest film is in the centre; whilst in these rings, obtained by chemical action, it exists at the circumference, as is the case with the colored rings of Nobili. In watching the formation of these phenomena, at first are seen two or three very small circles, which appear almost as soon as the iodine and the metal are placed in contact with each other; as the experiment continues, the circumference of these circles become gradually greater; whilst the external colors extend themselves over a great space, those of the centre grow fainter; red and green now only remain visible, and these at last, when the film has attained a certain thickness, in their turn also give place to a dull coating of brown.

The formation of these rings evidently depends on the vaporization of the iodine from the solid nucleus. The variety in color and extent of these zones is caused by the difference between the strength of the vapor at the centre and the circumference of the iodic atmosphere whilst expanding over so large a surface. In the metal thus combining with the vapor, we have to consider,—1, the force of the vapor of

different distances from the centre; 2, the obstacle which a film of iodine, once formed, opposes to any further action between the iodine and the metal.

This experiment may be varied in different ways: two pieces of iodine of about the same size, placed at a small distance from each other on a silver plate, form separate colored circles, until these come in contact at their circumferences, when the two systems will slowly coalesce and produce one common outline of the form of an ellipsis.

As the colors formed on various metals by the above-mentioned agents are very similar to one another, it may be sufficient to examine in particular those produced on silver by iodine.

The external film of the iodide of silver rings, which corresponds to the central black spot in those of Newton, is completely invisible, it being impossible to perceive any difference between the parts so covered and those where the metal is intact; but by exposing half the plate to the influence of light, whilst the other part remains covered, the silver is then found darkened far beyond the limits of the external gold-colored zone, where previously the surface was perfectly clear. The dark film thus rendered apparent is now liable to be rubbed off by the slightest friction, whereas before it was very adherent to the subjacent surface. The first zone is of a pale gold color, which assumes a deeper tint as the thickness of the film increases: the second zone is blue, the third white: after these appear the different colors of the spectrum in regular succession, as in the films studied by Newton and others, viz. yellow, orange, red, blue green, yellow &c.

The presence of the golden-colored zone in the place mentioned is worthy of remark, as in the tables of Newton of the colors presented by films of various thicknesses, the blue is stated as immediately following the black. The same gold film is the first which appears on most metals when their surface is attacked in this manner. Chlorine and bromine on silver; oxygen on steel; chlorine and bromine on titanium, bismuth, &c., commence their colors in the same way. Copper, however, is in one respect an exception, this metal first becoming of a dark red, which increases to a ruddy brown and then changes into blue. This deviation is fully accounted for by the color of the copper itself. With this single particularity, this metal undergoes the same alterations as the others.

The action of light on the different colors of the iodide of silver is very interesting: the most correct way of studying this is to protect one half of a system of colored rings by an opaque screen, while the other half is exposed for a short time to the influence of the solar rays. The golden zone undergoes the greatest change; at first it grows darker, then red, and at length is converted into a beautiful green. The blue film, which comes next in thickness, suffers considerable alteration in its tint,

assuming a much deeper and more brilliant shade; the rest of the colors appear to be similarly affected by the action of light, though to a very slight degree, acquiring a trifling accession in their brilliancy. It has already been remarked that light destroys the adherence of the external invisible film; the same thing obtains with the second or gold-colored film, which turns green, *but only to a certain depth* of the film, as may be proved by slightly rubbing the part thus altered; the green color is then seen to disappear, and beneath the pulverulent portion thus removed is found the gold color, having almost the same appearance as before the plate had been exposed. As this experiment may be repeated several times with the same results, it shows to how inconceivably small a depth the light has acted to produce this effect.

To ascertain what would take place on augmenting the thickness of the portion turned green, and the adherence of which was destroyed, a piece of iodine was placed on the plate so that its vapor, by expanding, might arrive upon the green, at the same time the whole being kept from the light; the result was that the additional film combined with the one already existing, producing a blue, being the color which would have resulted by the combination of the unaltered yellow films. I have found no chemical substance possessing the power of arresting, or in any way influencing these changes of color; strong acids, provided they do not attack the silver—for then, of course, the experiment would be destroyed,—and alkalies in concentrated solution, allow the action of light to go on as usual. The hyposulphate of soda, and ammonia in solution have no longer the power of dissolving the green film as they had before the action of light.

When the plate is left still longer exposed, after the changes above stated have taken place, the colors become more faint, and within the zone of green a white cloudy film is caused by the light, which, as it increases, veils the spectral colors beneath.

The knowledge we at present possess in chemistry of the affinities with which different bodies are endowed for combining with each other is but very imperfect, and the causes which complicate most chemical phenomena are so numerous, that it is scarcely possible to compare any two chemical actions to each other. Most of the facts upon which chemical science is founded, are acquired either by bringing the two bodies destined to act on each other into contact by dissolving them in a liquid, or by subjecting them to a temperature more or less elevated.

In the first of these methods, we are so far from being able to calculate the force of the chemical powers called into play, that Berthollet was induced to deny the existence of chemical power in the various phenomena of solution and precipitation of saline substances, and according to him what is called insolubility in a body is merely the result of its strength of cohesion, an entirely physical property.

When the intervention of caloric is required, the effects are still more complicated, as they vary according to the intensity of the heat employed, and the time its action is exerted; besides, the chemical action, when it does take place, is frequently so instantaneous that it is impossible in our present state of science to imagine any means by which it might be measured. In the combination of the three bodies, iodine, bromine and chlorine, with the metals, however, most of these objections cease to exist, or may be easily avoided. As their vapors combine with the metallic surfaces at the ordinary temperature, they are all of them in the same circumstances in that respect; and if the temperature should be required more elevated, the gasiform state of these substances, iodine not excepted, enables us to submit the metals to be experimented upon all at the same time to the same influence. If, therefore, it were possible to reduce the metallic substances into fine powders the particles of which were of the same dimensions, by acting upon them with either of these vapors, an idea might be formed of the affinities which produce their binary compounds by the increased weight acquired by the powders in this process; but the difference which exists in the physical properties of the various metals would preclude the possibility of any near approach to accuracy in this mode of proceeding; but by acting on the polished metallic surfaces, as in the preceding experiments, all the advantages offered by the process with the powders are included, whilst several of the difficulties are removed.

As the film of the compound augments, it undergoes the various changes of color which take place in all transparent films, thus affording a means of ascertaining the absolute thickness obtained in different circumstances, when it would be difficult to detect the slightest difference in weight by means of the most delicate balance. The depth of this coating may be ascertained when either the index of refraction of the compound itself is known, or if the angle of polarized light is given by means of the law discovered by Sir David Brewster, between the tangent of the angle of polarization, and the index of refraction. The most convenient way which occurred to me of performing these experiments, was the employment of a bell-glass within which some iodine is fixed at the top; this apparatus being placed over the metal to be acted on, the experiment may be watched in all its progress, and the action can be retarded or accelerated at pleasure by varying the interval of the iodine from the metal, or by interposing at some distance from its surface a disc of paper so as to cause the vapors of iodine to pass through it. Bromine may be made use of likewise by pouring a few drops of it over some carded cotton, and using it in a similar manner with the iodine. In respect to chlorine, it is most convenient to disengage it slowly by dropping a little sulphuric acid upon some chlorinated lime.

In illustration of the objects of this mode of experimenting, I will aduce some of the results it has given me with various metals. Some of the experiments below were performed before I had the idea of watching the progress of the combination through

a transparent medium; they are therefore less exact than they might otherwise have been: but I have preferred stating them as I had inserted them in my note-book before I had conceived any idea as to their probable utility in the elucidation of chemical affinity, and when I intended them for other purposes, which I shall hereafter explain.

Iodine with Silver and Copper.

- 1st change. Silver—pale gold.
" Copper—assumes a darker red.
" Silver—blue.
2nd do. Copper—blue.
" Silver—white.
3rd do. Copper—white.
" Silver—yellow.
4th do. Copper—yellow more extended than on the silver.
" Silver—Orange.
5th do. Copper—Red.
" Silver—blue, bluish-red.
" Copper—red, with a tinge of green on some parts.
" Silver—greenish blue.
" Copper—red, tinged with green.
" Silver—green.
" Copper—orange.
" Silver—yellowish green.
" Copper—orange tending to red.
" Silver—yellowish green.
" Copper—orange-red.
" Silver—red.
" Copper—dull green.
" Silver—red.
" Copper—green.
" Silver—deep green.
" Copper—dull red.

Bromine with Silver and Copper.

- 5th change. Copper—sensibly darkened.
" Silver—unchanged.
" Copper—deep red.
" Silver—unchanged.
" Copper—red, blue.

- " Silver—pale gold.
- " Copper—white, orange of the 2d order.
- " Silver—yellow.
- " Copper—green of the 1st order, red 3rd order.
- " Silver—blue.

Chlorine with Silver and Copper.

The affinity of chlorine with silver is much inferior to that which it possesses for copper.

Iodine with Titanium.

Iodine at the common temperature has no action upon this metal.

Bromine with Titanium.

Bromine, when the surface of this substance is perfectly dry, has no more action upon it than iodine; but if it have a slight coating of moisture, as is formed by merely condensing on it the vapor of the breath, the colored films are formed without difficulty by the vapors of bromine. Their appearance is the same as those of the iodide of silver, viz. gold, deep gold, blue, white, yellow, orange, red, &c.

Chlorine with Titanium and Copper.

Titanium has a stronger affinity than it has for either of the preceding vapors. The combination takes place when the metallic surface is either dry or moist.

Copper—much reddened.

Titanium—not affected.

Copper—passed through several of the spectral orders of red and green until it arrived at almost its last changes of colors.

Titanium under the same action received a dull film, which viewed obliquely showed red, green, yellow.

Silver, exposed to the same influence as the two former, had yellow in the centre and blue more externally.

Iodine with Bismuth and Silver.

Silver—pale gold.

Bismuth—some parts yellow, others not attacked.

Silver—blue, white, yellow-orange.

Bismuth—blue, yellow, orange.

In the action of iodine on bismuth, the influence of the physical condition of metallic surface is very manifest. The crystalline texture of this metal may be perceived, and the difference of its hardness admits, to a certain point, of being measured by the difference of the color of the films that are formed on various points; while most parts are yellow, there exist others of an angular outline which remain still unattacked; the same difference is remarked in the other stages of the combination.

Iodine with Mercury.

It is impossible to estimate the affinity between mercury and iodine by means of the colored films, because, on combining, these two substances merely cause a dirty white appearance on the surface of the latter. Their combining affinity appears to be considerable, for when exposed together with silver the action produced with both was red at the edges, little altered in color; on the rest of its surface a dull white film, in the midst of which were seen several dark spots, where the metal was apparently unaltered.

Bromine with Mercury and Copper.

- 1st. Mercury—gold color.
 - " Copper slightly darkened.
- 2nd. Mercury—blue.
 - " Copper dark red.
- 3rd. Mercury—green on some parts.
 - " Copper white.

After this the copper underwent its usual changes of color on prolonging the action of the vapor of bromine, but the color of the mercury suffered no further change.

Chlorine with Mercury and Copper.

- Mercury—a slight film.
- Copper no alteration of color.
- Mercury—deep gold color.
- Copper deep red on some parts, blue on others.
- Mercury—red tinged with blue.
- Copper blue, white.
- Mercury—blue.
- Copper same as before.

"With respect to the bromide and chloride of mercury, it is necessary to view them obliquely in order to perceive all the changes of color they undergo; for if looked at perpendicularly, there is seen on both a dull uneven film of white which

reflects none of the above colors: consequently, to avoid any error, the copper must be inspected under the same angle.

Bromine with Bismuth and Silver.

Silver—pale gold.

Bismuth—not apparently changed.

Silver—deep gold, blue.

Bismuth—yellow, blue.

Silver—blue, yellow.

Bismuth—dull colorless film.

Chlorine with Bismuth and Silver.

Bismuth is slowly attacked with chlorine gas, much in the same way as with iodine and bromine in vapor.

Bromine with Lead.

At the common temperature neither bromine nor chlorine forms colored films upon this metal, which it is very difficult besides to bring to any high state of polish on account of its softness. But when lead is heated, as over the flame of a spirit-lamp, the vapors of bromine then form very fine colored films, which are in succession gold, deep blue, &c.

Iodine with Iron.

These two may be made to form colored films when combined rapidly together, but generally a dull coating without any spectral color is obtained, on account of the deliquescence of that salt.

Until we know the index of refraction of the different films enumerated, it would be impossible to give a correct table of the combining powers in the experiments that have been detailed; nor is the table of the relative thickness of transparent plates as it has been transmitted to us by Newton, sufficient in the present instance, if any great degree of precision be required. Besides these objections, it is necessary before leaving this subject to pass in review several others inseparable from the mode of performing the experiments themselves. The principal circumstances complicating these experiments and liable to vary in different observations, are,—

First, the hardness of the metal acted upon; 2ndly, the obstacle opposed to the continuation of chemical action by the inert film formed upon the metal; 3rdly, the force of the vapors that attack the metal. The influence of the texture of the metallic surface on chemical action is most evident when bismuth is the metal employed. Here the chemical action may be seen to commence on small isolated portions of the surface, which have already assumed a deep gold color, before other parts are in the

least changed, from the natural appearance of the metal. To determine how far this might influence the formation of the iodide of silver, a silver coin was exposed to iodine with a piece of pure silver; as the former was so much the harder of the two, it was naturally supposed that the chemical action would be slower in exerting itself on it than on the latter. This, however, was not the case, as may be seen by the following statement of the result of the experiment:

Silver coin—pale gold color.
Pure silver—pale gold.
Silver coin—deep gold.
Pure silver—deep gold.
Silver coin—light blue.
Pure silver—light blue.
Silver coin—yellow.
Pure silver—blue, white, yellow not visible.
Silver coin—yellow, red at edges.
Pure silver—yellow, no red edges.
Silver coin—red, blue at edges.
Pure silver—yellow, no red apparent.

The intensity of the resistance offered by the different films of iodide of silver to a continuation of the chemical combination, may be determined by noting the moment at which the various spectral tints make their appearance.

Color of the film of iodide of silver.	
'	"
0 50	—beginning to darken.
2 0	—pale gold.
4 40	—deep gold.
6 40	—orange blue.
7 30	—blue.
9 30	—light blue.
11 30	—commencement of yellow.
18 30	—orange red.
20 15	—blue.
22 55	—deep blue.
24 40	—green.
28 0	—yellowish green.
30 25	—ruddy brown.
40 10	—green.
46 30	—green.

50 10—red.

53 15—green.

By comparing the thickness of the colors with the space of time required for their production, it will be found, however imperfect the table given by Newton may be when applied to this subject, that towards the end of the experiment above given, the chemical combination is retarded by the presence of the inert film, and that to obtain the same thickness of film as at the commencement, about double the time is required.

The third cause of error may be avoided by operating with vapors of about the same force. In those described, the average time employed in passing to the maximum was generally about half an hour; if that were not taken into consideration, different results might be obtained.

In regard to chlorine, there exists another cause of complication—the affinity which it possesses for water; for when disengaged in the ordinary manner, chlorine carries with it a certain quantity of water which may very much alter the results of the experiment.—*Philosophical Magazine*, 1842.



THE AMERICAN ELECTRIC TELEGRAPH.

In each of the countries of Europe where the Electric Telegraph is established, viz.: Britain, France, Germany, and Russia, the people seem anxious to claim for one or more of their own countrymen some merit in connection with the invention. The English speak of Wheatstone and Bain; the French of Le Sage; the Germans of Steinheil and Siemen; and the Russians of Schilling or Jacobi. Most of these inventors have their agents or representatives in different countries, ready on every suitable occasion to speak or write in behalf of their principal.

The inventor of the American Electric Telegraph has, we believe, no agent in Europe; and no person there who has any special interest in promoting the introduction of his system. The adoption of it by the Prussian government for great distances, as decidedly the best which it could obtain, after advertising extensively in Europe, is, therefore, the more remarkable; and especially, as the only knowledge which the Prussians possess of the American system seems to have been derived from one imperfectly acquainted with its powers; for we can account in no other way for the fact that Siemen's needle apparatus is adopted for short distances; Morse's instruments being clearly superior to Siemen's, or to any other needle apparatus, for short as well as for long distances; since they are much more rapid in their motion, and have, besides, the advantage of making a permanent record on paper, instead of merely indicating to the eye by an evanescent sign. No one using the telegraph in America, would think of employing a needle apparatus in any of its operations.

In addition to the evidence of the high estimation of the American Telegraph by Europeans, furnished by this adoption of it in Prussia, we learn also from an elaborate work on Telegraphs by Dr. H. Schellen, recently published in Dusseldorf, that it is now introduced into other countries on that continent. Dr. S. says: "Of late the Morse Telegraphs are much used in Europe, viz: in England; between Hamburg and Cuxhaven; between Bremen and Bremerhaven; in Prussia, Austria, Bavaria and Hanover." Dr. S. might have added also that it is used in Turkey. Among the advantages of the American Telegraph, Dr. S. says, is its "quickness in making and marking the signs upon paper. * * * *It records under the hand of a skilful operator 100 to 120 letters, in the same time in which the best needle telegraphs are able to indicate 20.*"

While the superiority of the American Electric Telegraph to all others is thus acknowledged by Europeans, they do not seem to be aware that the date of the American invention is also prior in point of time to that of any of their own electric telegraphs. They very naturally regard the date of the patent as the date of the

invention, because in every country in Europe the patent is given to the person who first promulgates the invention in that country; and it is not safe, therefore, in Europe, for an inventor to permit others to know any thing of his invention until he has patented it. But in America, where the law permits no one to take out a patent but the inventor, he is perfectly safe in communicating the knowledge of what he has done, and even in publicly exhibiting his invention, before he takes out his patent, provided he secures himself by a caveat, and does not offer it for sale, or permit it to go into common use.

The inventor of the American Electric Telegraph, it is well known, exhibited his instruments in operation in the N. Y. University, in the presence of hundreds of our citizens in September, 1837, and privately to his friends at various times as far back as November 1835, although his first patent was not secured until 1838. No one claims for the inventions of either Wheatstone or Steinheil a date prior to 1837; and when European writers on the telegraph come to understand our laws, and the error into which they have fallen, by confounding the date of the American invention with the patent, they will no doubt do our country the justice to admit its claims to priority as well as to *superiority*.—*N. Y. Observer*.



IRON PAVEMENT.—Iron is daily coming into more general use for almost every purpose. A letter from Paris, of a late date says:

"A new pavement, to upset the Macadam and other inventions of the kind, has been proposed by Mr. Tobard, who intends paving, in this way, the streets and boulevards of Paris. This gentlemen has proved, by figures, that melting iron is only worth 11 francs in Paris, 7 francs in Belgium, and 4½ francs by 100 kilogrammes in England, whilst the stone costs 25 francs in London, 15 francs in Paris, and 8 and 10 francs in Belgium. This new mode of pavement will be grooved, in order not to become slippery, and it is said that the electricity occasioned by the rolling of the carriages will prevent rust. Here is a new field open to industry."



CLAUDET'S SPECIFICATION.

SEALED 21ST NOVEMBER, 1843.

WE give the following as copied at the Patent Office in London. Many of our Daguerreotypists have enquired of us in relation to the patents on the Daguerreotype in England.

TO AUTOINE JEAN CLAUDET, of High Holborn, in the county of Middlesex, glass merchant, for an invention of improvements in the process and means of obtaining the representation of objects of nature and art,—being a communication.

[Sealed 21st November, 1843.]

These improvements consist in rendering the Daguerreotype picture susceptible of producing, by printing, a great number of proofs or copies; thereby transforming it into a complete engraved plate.

The process is established on the following facts, which have come to the knowledge of the inventor:—

1st. A mixed acid, composed of water, nitric acid, nitrate of potassa, and common salt, in certain proportions, being poured upon a Daguerreotype picture, attacks the pure silver, forming a chloride of that metal, and does not effect the white parts, which are produced by the mercury; but this action does not continue long. Then, by a treatment with ammonia (ammonia containing already chloride of silver in solution, is preferable for this operation), the chloride of silver is dissolved, and washed off, and the metal being again in its naked state, or cleansed from the chloride, it can be attacked afresh by the same acid. This acid acts better warm than cold.

2d. As all metallic surfaces are soon covered (when exposed to the atmosphere) with greasy or resinous matters, it is necessary, in order that the action of the acid upon the pure silver should have its full effect, for the surface to be perfectly purified; this is effected by the employment of alcohol and caustic potash.

3d. When a Daguerreotype picture is submitted to the effects of a boiling concentrated solution of caustic potash, before being attacked by the acid, the state of its surface is so modified that the acid spares or leaves, in the part which it attacks, a great number of prints, which form the grain of the engraving.

4th. When the effects of the acid are not sufficient, or, in other words, it is not bitten deep enough, the effect is increased by the following process:—Ink the plate as copper-plate printers do, but with a siccative ink; when the ink is sufficiently dry, polish the white parts of the plate, and gild it by the electrotype process; then wash it with warm caustic potash, and bite it in with an acid, which will not attack the gold, but only the metal in those parts which having been protected by the ink, have not received the coating of gold. By these means the engraving is completed, as by the acid alone it is not generally bitten in deep enough.

5th. To protect the plate from the effects of wear, produced by the operation of printing, the following process is employed: The surface of the plate is covered with a thin coating of copper, by the electrotype process, before submitting it to the operation of printing; and when that pellicle or coating of copper begins to show signs of wear, it must be removed altogether, by plunging the plate in ammonia, or in a weak acid, which, by electro-chemical action, will dissolve the copper, without effecting the metal under it; the plate is then coppered again, by the same means, and is then ready for producing a further number of impressions. This re-coating operation may be repeated as many times as may be required. The following is the description of the whole process, which is divided into two parts, consisting of a preparatory and printing process.

Preparatory Engraving.—For this operation which is the most delicate, it is necessary to have—1. A saturated solution of caustic potash. 2. Pure nitric acid at 36° of the barometer of Beaumé (spec. grav. 1·333). 3. A solution of nitrate of potassa, composed of 100 parts of water, and 5 parts of nitrate, by weight. 4. A solution of common salt, composed of water 100 parts, and salt 10 parts by weight. 5. A weak solution of ammoniacal chloride of silver, with an access of ammonia. The ammoniacal chloride of silver must be diluted with 15 or 20 parts of pure water. In the description of the process, this solution will be called ammoniacal chloride of silver. 6. A weak solution of ammonia, containing 4 or 5 thousandths of liquid ammonia. This solution will be called ammoniacal water. 7. A weak solution of caustic potash, containing 4 or 5 thousandths of the saturated solution, which will be called alkaline water. 8. A solution composed of water 4 parts, saturated solution of potash 2 parts, alcohol 1 part, all in volume. This solution will be called alcoholized potash. 9. Acidulated water, composed of water 100 parts, and nitric acid 2 parts, in volume. Besides, it is necessary to have three capsulæ or dishes, made of porcelain, large enough to contain the plate, and covered with an air-tight piece of ground plate glass, and two or three more capsulæ, which do not require to be covered; two or three glass funnels, to wash the plate; and two or three glass holders, in the shape of a spoon or shovel, by which the plate is supported when put in and taken out of the solution, without touching it with the fingers.

The Daguerreotype plate is submitted to the engraving process, after having been washed in the hyposulphate of soda, and afterwards in distilled water.

First process for biting in or engraving the plate.—The following solutions must be put in the capsulæ, in sufficient quantity, so as to entirely cover the plate:—1. Acidulated water. 2. Alkaline water. 3. Alcoholized potash, in covered capsulæ. 4. Caustic potash, in covered capsulæ. 5. Distilled water.

The plate being then put upon the glass holder or spoon, is plunged in the acidulated water, and agitated during a few seconds, then put into a glass funnel, and washed with distilled water. It is taken again with the glass spoon, and plunged in the capsulæ containing alcoholized potash. This capsulæ is covered with its glass cover, and then heated, by means of a spirit lamp, to about 140° Fahrenheit. The plate must remain in the capsulæ half an hour, during which the solution is heated now and then, and agitated. During that time the following acid solution, which will be called *nomal acid*, must be prepared: it is composed as follows:—Water 600 parts, nitric acid 45 parts, solution of nitrate of potassa 12 parts, solution of common salt 45 parts. These proportions are in volume. The nomal acid must be poured in a capsulæ, covered with its glass cover, and a sufficient quantity must be kept in the bottle.

When the plate has been immersed in the alcoholized potash during half an hour, it is taken out of the solution by means of the glass holder, and immediately plunged in the alkaline water, and agitated pretty strongly; from thence it is put in distilled water (A). This being done, the plate is plunged in the acidulated water, and moved about therein for a few seconds: it is then put into the nomal acid. When the plate has been immersed a few seconds in the acid, it is taken out by means of the glass holder, taking care to keep it as much as possible covered with the solution, and it is immediately placed horizontally upon a stand, and as much acid as the plate can hold is poured upon it from the bottle; it is then heated with a spirit lamp, but without attaining the boiling point. During this operation it is better to stir or move about the acid on the plate by pumping it, and ejecting it again, by means of a pipette or glass syringe; after two or three minutes the acid is thrown away, the plate is put in the glass funnel, and then well washed with water, and afterwards with distilled water (B).

Thus, without letting the plate dry, it is put upon the fingers of the left hand, and with the right hand some ammoniacal chloride of silver, which is moved about the surface by balancing the hand is poured upon it; the solution is renewed until the chloride, formed by the action of acid, is dissolved; the plate is then washed by pouring upon it a large quantity of ammoniacal water, and afterwards some distilled water. (C.)

Without allowing the plate to dry, it is then put in the caustic potash, and the capsula being then placed upon the stand, the potash is heated up to the boiling point; it is then left to cool (D); and beginning again the operations described from A, to D, a second biting is obtained; and by repeating again the operations described in A, and B, a third biting is produced. The plate is then dried; in this state the black parts of the plate are filled with chloride of silver.

The plate is then polished until the white parts are perfectly pure and bright. This polishing is soon done with cotton and ('ponce') (pumice stone); afterwards, the chloride of silver, filling the black parts, is cleansed by the means described in B, and C. The plate is dried, but before drying, it is well to rub the plate slightly with the finger, in order to take off from the black parts any remains of an insoluble body, which generally remains on it. The preparatory engraving is then finished, and the plate has the appearance of a very delicate aquatint engraved plate, not very deeply bitten in.

Nevertheless, if the operation has been well managed, and has been successful, it is deep enough to allow the printing of a considerable number of copies.

Note.—Sometimes, instead of treating the plate with the boiling potash in the capsula, a similar result may be obtained by placing the plate upon the stand, covering it with the solution, and heating it by means of a spirit lamp, until, by evaporation the potash becomes in a state of ignited fusion. By this means the grain is finer, but the white parts are more liable to be attacked.

Last operation of biting in: This operation requires some of the re-agents, before-named, and also,

1. A siccative ink, made of linseed oil, rendered very siccative by boiling it sufficiently with litharge; it may be thickened with calcined lampblack.

2. An electrotpe apparatus, and some solutions of it to gild, and copper the plate.

Means of operating: The plate must be inked as copper-plate printers do, taking care to clean off the white parts more perfectly than usual; the plate is then to be placed in a room sufficiently warm, until the ink is well dried, which requires more or less time, according to the nature of the oil employed. The drying of the oil may be hastened by heating the plate upon the stand with the lamp, but the slow process is more perfect and certain.

When the ink is well dried, the white parts are cleaned again, by polishing the plate with cotton and ponce, or any other polishing powder; a ball of cotton, or any other matter covered over with a thin piece of caoutchouc or skin, can be used for

this purpose. When polished the plate is ready to receive the electro-chemical coating of gold, which will protect the white parts.

Gilding.—The gilding is obtained by any of the various processes of electrotyping that are known. The only indispensable condition is, that the surface obtained by the precipitation must not be liable to be attacked by the weak acid; a solution answering this purpose is made of 10 parts, (by weight), of ferrocyanide of potassium; 1 part of chloride of gold, and 1,000 parts of water, used with a galvanic battery. During the gilding the plate must be turned in several positions, in order to regulate the metallic deposit. In some cases the gilding may be made more perfect, if the plate is covered with a thin coating of mercury before putting in the gilding solution.

When the plate is gilded, it must be heated with the boiling caustic potash, by the process already indicated for the preparatory engraving, in order to cleanse it from all the dried oil or ink, which fills the hollow. The plate is then washed and dried, and when the oil employed has been thickened with the lampblack, the surface of the plate is rubbed with crumbs of bread, in order to cleanse and take off the black remaining; then, the white parts being covered and protected by varnish not liable to be attacked, and the black parts being uncovered and clean, the plate can be bitten in by aqua-fortis, according to the ordinary process used by engravers.

This operation must be used upon the stand, and not by immersing the plate in the solution.

Before this biting in, if the preparatory engraving has not succeeded well, and the plate still wants a sufficient grain, it can be given by the various processes of aquatint engraving.

Before submitting the plate to the operation of printing, in order to insure an unlimited number of copies, it is necessary, as before stated, to protect it by a slight coating of copper, which is obtained by the electrotype process; otherwise the printing would soon wear the plate. This coating must be kept very thin, but the fineness of the engraving, and the polish of the white parts, should be destroyed. In this state the plate can be delivered to the printer.

After a certain number of impressions have been obtained, it will be perceived that the coating of copper is worn in some places; then this coating must be removed, and a fresh one applied in its place. For this purpose, the plate must be purified and cleansed by warm potash, and plunged in a weak acid, composed as follows:—Water, 600 parts; nitric acid, 50 parts; nitrous acid of engravers, 5 parts; all in volume. This acid will dissolve the coating of copper, and the plate being coppered again, by the same means as before, may be again submitted to the operation of printing; and as nothing can prevent the success of a repetition of the same operation, any number of

impressions may be obtained. The coating of copper can also be removed by caustic ammonia. The Daguerreotype plate engraved by this process, may also be reproduced and multiplied by the electrotype process, the same as any other engraved plate.

The essential points of this process, which constitute the present invention, consist,—

First,—in the discovery and employment of certain properties of a mixture composed of nitric acid, nitrous acid, and hydrochloric acid, in detrimental or fixed proportions. The two last mentioned acids may be employed either in a free state, or combined with alkaline or other bases. This mixed acid has the property of biting the pure silver, which forms the black parts of the Daguerreotype picture, without attacking the white parts formed by the amalgam of mercury.

The result of the action of the biting is, to form on the black part of the picture an insoluble chloride of silver; and this chloride of silver, which, when formed, stops the action of the acid, is dissolved by the ammonia, which allows the biting to continue.

Secondly,—In the discovery of certain properties of a warm solution of caustic potash, and in the employment of the said solution, by which the mercury forming the picture is better and deeper amalgamated with the silver under it, so that many imperceptible points of the amalgam are affected in such a manner that the action has no action upon them.

Thirdly,—In the discovery and employment of a process which produces a grain favourable to the engraving, by which the biting on the plate is rendered deeper. This is effected by filling the parts engraved with a siccative ink, or any other substance, and then gilding the plate by the electrotype process: the gold is not deposited on the parts protected by the ink. When the plate is gilded, the ink is cleansed by the caustic potash, and the plate may be submitted to the effect of an acid, which does not attack the coating of gold, but bites only on the silver in the parts already engraved by the first operation.

Fourthly,—In the employment of a process by which the plate is protected from the wear of the printing operation. This is effected by covering the plate before printing, with a slight coating of copper; by the electrotype process, and when the coating begins to wear, by printing, it is removed by a weak acid or by ammonia, which dissolves the copper without effecting the silver under it. The plate is coppered again, and after another printing, the same operation is repeated, so that a considerable number of copies may be printed without injury to the engraving.

Enrolled in the Pitty Bag Office, May, 1844.



INTERESTING EXPERIMENT WITH STRYCHNIA.

An interesting experiment, illustrative of the poisonous effects of strychnia, was recently made by Professor Agassiz, at Cambridge. The subject was a large black bear, about eighteen months old. The animal was taken when young, and had been kept in captivity for a considerable period. Professor Agassiz being desirous of killing it for the purpose of dissection, about three grains of strychnia were administered. The poison, though extremely bitter was readily swallowed. At the expiration of ten minutes, no effect having been produced, a second dose of about the same quantity was also inclosed in a biscuit and offered. The cunning animal broke open and swallowed the biscuit, but rejected the poison. The first portion, however, had proved efficacious, and in exactly fifteen minutes from the time when first administered, the animal was seized with terrible convulsions, and soon died.

The whole time which elapsed between the taking of the poison and the death of the animal, did not exceed twenty-five minutes. In order to alleviate its sufferings and hasten death, a quantity of hydrocyanic acid was poured upon the nose and mouth of the bear. It did not, however, produce any sensible effect, and was not apparently taken into the system, as the animal at the time was nearly dead. But the subsequent effects of the poison were most remarkable. Although the bear, at the time of death, was in perfect health and strength, twenty-four hours had not elapsed before the body was in an advanced stage of decomposition. Indeed the appearances indicated that the animal had been dead for nearly two months. The interior of the body, when opened about twenty hours after death, still retained its warmth in a considerable degree, while an offensive gas issued from every pore. The blood had not coagulated, the spinal marrow and nerves were in a semi-fluid state, and the flesh had assumed a leaden-gray color. The hair of the hide readily came out on being slightly pulled. No smell of the hydrocyanic acid could be perceived.

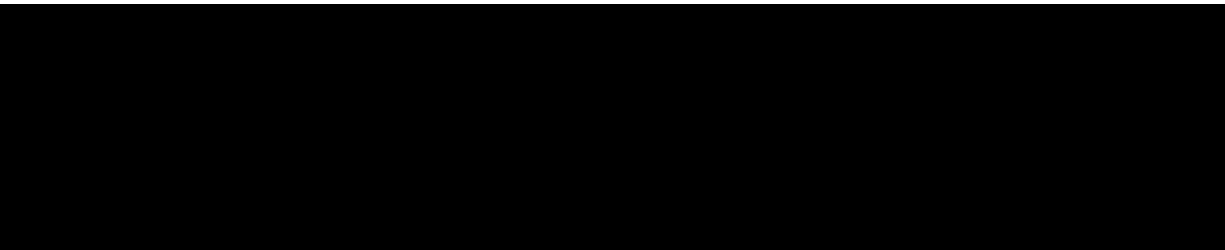
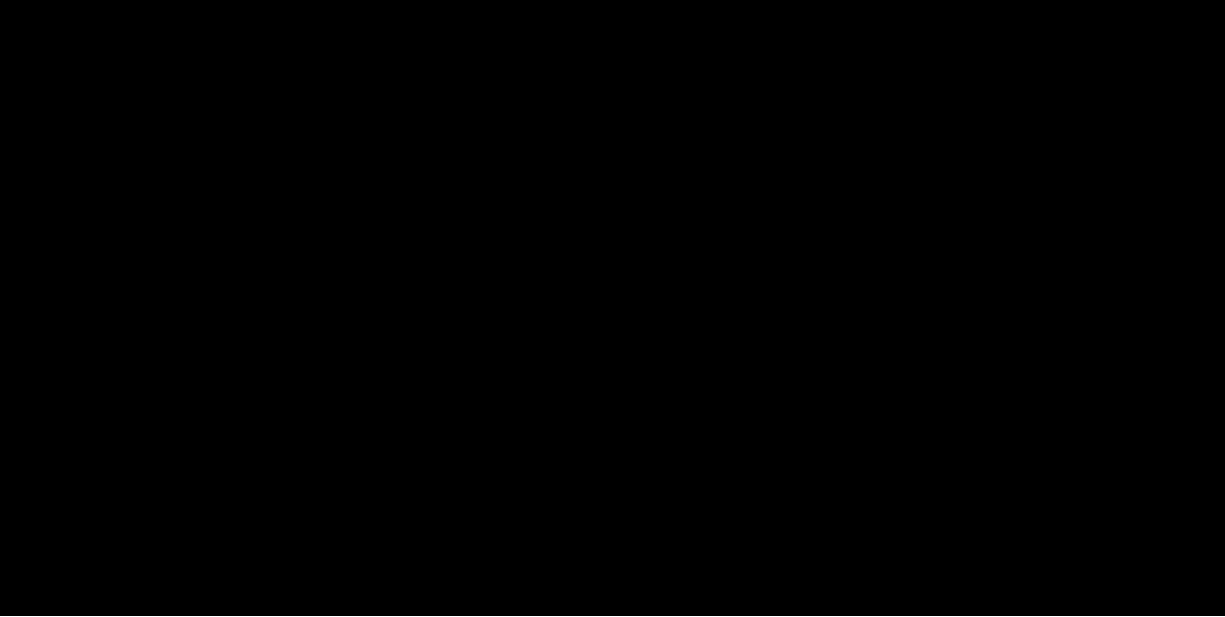
The origin of this singular and speedy decomposition is not fully known, though it is supposed to be due to the agency of the hydrocyanic acid. A chemical examination of the muscle, brain, nerves, liver and kidneys is now going on at the Cambridge laboratory, under the direction of Professor Horsford. One singular fact connected with the spontaneous decomposition of these parts is, that they all yielded or disengaged hydrosulphuric acid gas, with the exception of the liver, which did not. —*Annual of Scientific Discovery.*



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THE DAGUERREIAN JOURNAL

NEW YORK, JUNE 1, 1851.

The metal which the Daguerreotypists have most to deal with is silver. To obtain plates of this perfectly clean and keep them so is one of the most difficult parts of his art; in order to accomplish it he is obliged to have resource to circuitous mechanical processes, or to some chemical means whereby the old surface may be removed, or a new surface deposited. He uses tripoli and rotten stone, rouge and buffing, fatty matters, alcohol, ammonia, and nitric acid, for the first purpose; and electro-plating for the second. The operator must not mistake the object he has in view. By all his mechanical operations, where friction is concerned, he obtains a *polished* surface; but it is not to *polish* the plate that he uses the buff, and the other materials; the object is to clean the plate, and the *polish* of the surface follows as a natural result. That a polished surface is not necessary for the formation of an image is evident from photography on paper, and Hyalotype. In these cases the surfaces do not reflect light; they are not polished; they are merely clean. The fact of a plate freshly taken out of a cell where silver has been deposited on it receiving beautiful images, is another proof that polish is unnecessary for success. What is wanted is a surface of pure silver, one not coated with the slightest trace of oxide, carbonate, or sulphuret of silver, or with any impurity derived by contact with foreign matter. The Daguerreotypist has, then, two objects in view.—1st, to get a clean surface on his plate; and 2nd, this once obtained, to keep it so without further alteration. Although silver is classed with gold and platinum, as one of those metals which do not tarnish or oxidize readily on exposure to the air, yet it is now well known in practice what a difference exists in the character of the picture brought out on a plate recently polished and one set aside even for a few hours. It is true that the atmosphere alone will not affect or oxidize a plate of polished silver, but that is only true of a very dry atmosphere; for an atmosphere which contains even a small amount of moisture in it, will in a short time dull the surface of the plate. In this instance the moisture is deposited first, and brings the air into closer contact with the surface of the metal, promoting the union of the latter with the oxygen of the air. This is exemplified every day in the case of iron, which does not rust in dry weather, but when the air begins to deposit its moisture it rapidly oxydizes. So is it with lead and zinc, but in a less degree. Pure water has no effect in tarnishing the surface of a silver plate, but if it contains even a very small

quantity of animal or vegetable matter it darkens the surface in consequence of the presence of sulphur. The ordinary tarnish of silver is due to the fastening upon the surface of sulfurous vapors, which are constantly floating in the air. We are scarcely yet in possession of sufficient facts to justify the statement that sun-light materially aids in this alteration. We do know that oxidizement in any substance goes on much more rapidly in sunshine than under any other circumstances. In the vegetable kingdom it is of hourly occurrence; but the different effects of shade and shine upon the oxidizement of plates of metal has not been minutely studied. However this may be, after a time the silver surface does get coated with a film, of partly sulphuret, partly oxide, and a complete layer of air, which so closely attaches itself to the plate that water when poured on it rolls off and does not wet the surface. It is this film, of air which mechanical friction and bulling are so serviceable in removing. The tarnish of a sulphuret is not so readily removed by friction as by ammonia, for the sulphuret of silver is, to some extent, soluble in ammonia; but as the sulphuret when present gives a yellowish brown tint to the surface, it is only then that it is needful, and even a weak solution of common salt or hydrodate of potash cleans the surface much quicker.

As nitric acid is the best solvent of silver, so is it the best agent to remove films of oxide and carbonate from the surface; it does so readily, effectually, and leaves a clean new surface of silver, similar to a plate newly prepared, or one which has received a new coating in the electrotype cell. The acid, however, must be very much diluted, otherwise its local action is so immediate as to eat down the surface of the plate faster than it spreads, producing inequalities,—even when left long on, although diluted, it produces roughness; hence its use has not been happy in many hands. To such, the longer, but less failing processes of mechanical action are to be recommended in preference.



FRIEND HUMPHREY:—

I have thought for some time past that I would furnish a few thoughts for the *Daguerreian Journal*, under the head of *General Rumors*. A fear of injuring the Daguerreian business, by giving an additional publicity to my discovery, for the present has kept me from this, as well as from furnishing several other articles.

A very few persons (for the honor of humanity, I am glad the number is small) have made statements which reflect seriously upon the integrity and motives of myself and friends. By means of certain legal processes, of which I am not wholly ignorant, I could do *these gentry* more simple justice than by dignifying them by a notice in our Journal, yet I bear them no ill will, and *for the present* shall act on the principle that it is better to suffer wrong than to engage in a quarrel, especially with

men who have so little regard for the feelings and reputation of their fellows, as to condemn, denounce, and accuse, without discrimination, and without the slightest knowledge of facts.

The "head and front of my offending" is this—that I *have*, by dint of unwearied pains and perseverance, succeeded in imprinting the colors of nature, in all their inimitable glory, upon the metallic surface. The process, when brought out, it is believed, by the best of judges, will be an astonishment to, the world, and a delight to our nation; and I have felt, from the first, that I am but an instrument in the hands of the Almighty, and am therefore bound in duty, as well as by inclination, to dispose of the secret in a way to accomplish the greatest amount of good. In this spirit I have frequently published my unalterable resolve that my process *shall never be monopolized, or used for the purposes of extortion*. If I had no other reason for this course, the fact that there are great numbers of worthy persons engaged in our art, who, like myself, are poor, this would be sufficient. Various plans have been suggested to me, for the best possible method of disposing of my discovery. Numerous gentlemen of high standing, have honored me with personal intercourse and friendly advice. I have even suffered the claims of courtesy to interfere with my experiments, and, I believe I have the good will and confidence of those who have honored me with their visits; but I have never swerved from my original determination, and have given no person the slightest reason to think otherwise.

In view of these *facts*, it is not strange that I should feel annoyed at the various rumors to which I refer, and some of the intimations which have been thrown out, I *feel* are but poor pay for my having nearly, if not quite, ruined my health, by laboring from morn till midnight for the good of others. More especially have I been *grieved* that *some of my best friends, and the best friends of the fraternity*, should be accused of having outwitted me, and drawn me into obligations to them which would compel me to pursue a grasping course. Now, it is true that certain gentlemen have, from the most honorable motives, made me very liberal offers of aid, and it is also true that my mind is about settled in regard to the parties with whom I shall act; but it is not true that I am under obligations to give notoriety to my *business affairs*, and it is certain that I shall not do so until I see fit. Neither is there the least foundation for the assertion that I am committed to any person. There confidence in me is not based on paper, but on a much surer foundation.

The process which I have been so fortunate as to discover, is *my property*. No man on earth has any claim, upon it, and I have a perfect right to dispose of it as I please, in case I do not interfere with the rights of others. This, I think, is the philosophy of *Professor Morse*, so admirably and clearly expressed in the last number of the Journal. This distinguished man recently favored me with a visit, and, in the course of the interview, shed much light on this subject for which I feel truly thankful.

My intention has been, and is, to *do right*, and I believe the majority approve of the course I have pursued thus far. The many marks of approbation I have received from eminent men, both in this profession and others, is truly encouraging to me in my poor health and arduous labors, and will never be forgotten. Those who seem to possess a different spirit, and who have apparently endeavored to hedge up my way by means of "rumors," "new discoveries," "threats," &c., have my forgiveness, and I trust that, in future, if they are not disposed to aid, they will do nothing to hinder me.

The brotherhood have nobly aided me, by purchasing my book. My sales are exceeding my most sanguine expectations, and are placing me on a footing to make me *feel that I am free*. All may be assured that I shall not use my freedom "as a cloak," but to serve their interests, of which both myself and friends hope to give them *full proof* hereafter.

Now that I have been compelled to close my doors against visitors, I would state that I am doing my utmost to bring out my process in a short time, and I will hereafter give, in the *Daguerreian Journal*, such notices of my success as will meet the wishes of the public, as well as the interest of our fraternity,

I am yours &c.,
L. L. HILL.

Westkill, Greene Co., N. Y.,
May 30, 1851.

☞ The foregoing letter was addressed to me privately, and as it contains some of Mr. Hill's views in relation to the various rumors afloat, I have thought it best to insert it for the public.

S. D. H.



We take pleasure in saying to our friends that, through the kindness of John Johnson, Esq., of this city, we have received many of the valuable letters and papers written by Mr. S. A. WOLCOTT, a former partner of the first named gentleman. Mr. W. is well and favorably known as one of our early and most successful investigators in the Photogenic Art. Many of the most valuable improvements had their origin in his hands. We were surprised to find so great an amount of information as is here presented.

Our Daguerreotypists of the present day may consider themselves fortunate, when looking back upon the difficulties experienced by those early engaged in putting a shadow on the silver plate. We find here many a mystery unravelled, and set forth in a light heretofore unexplained. As these letters, written by Mr. W. to his

partner, Mr. J., (at the time they were written) were many of them confidential, and some few contain business matter of a foreign character, it will require some little time to prepare them for the press in the order in which they were written. We shall make no alteration in the manuscript, and exclude such matter only as relates to business of a foreign character. They present a team of investigation and research worthy the first in the scientific circle.

We would here take occasion to speak of a curious specimen of our art, which we saw while recently spending an evening with Mr. Johnson. This specimen is a profile view of a gentleman, and if not the *first* likeness from life, it is said to claim age with the oldest Daguerreotype ever produced, and is no less interesting for its antiquity than for its size, it being on a plate *less than one quarter of an inch square*. To produce an impression on such a plate it would require the superior skill of the most successful operator of the present day. The outline is distinctly marked, the features remarkably well delineated, being in fact, one of the most interesting curiosities we have ever witnessed. The plate is cemented by means of Canada Balsam to a piece of plate glass about three-fourths of an inch square, and thus it has been preserved in its present state.

We saw many other specimens of Daguerreian antiquity, but as we intend to speak more fully of our evening visit at Mr. Johnson's, we will defer further notice for the present.

We would here say to such as may have Mr. Walcott's letters relative to the art, that we would consider ourselves especially favored if we could be allowed the perusal of them.



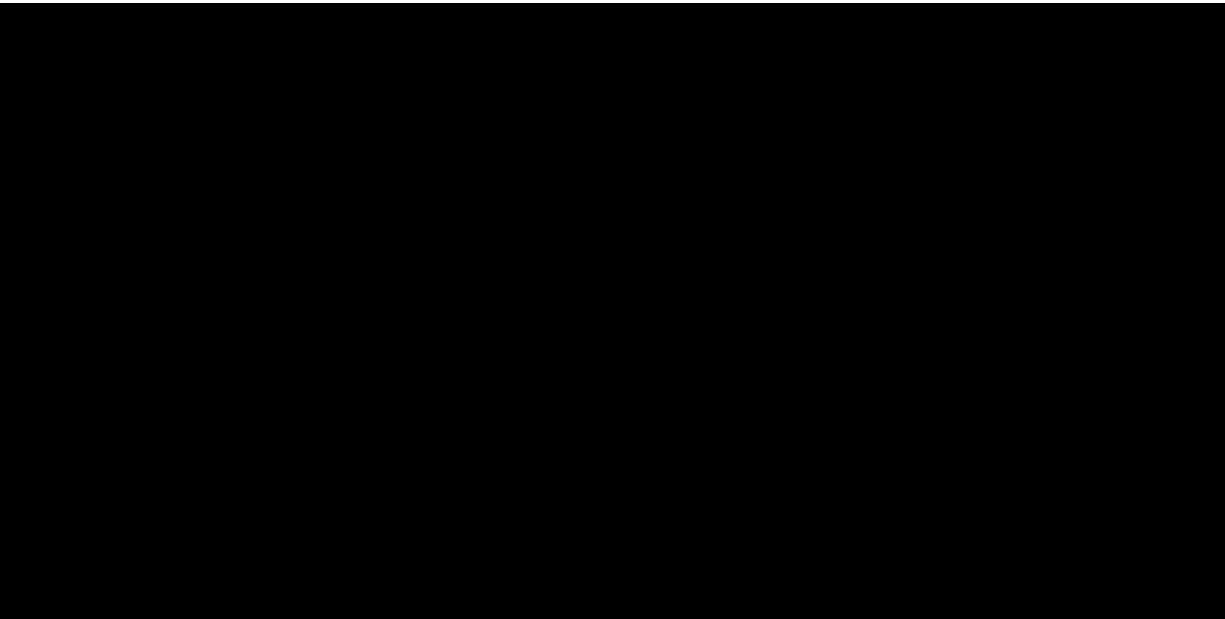
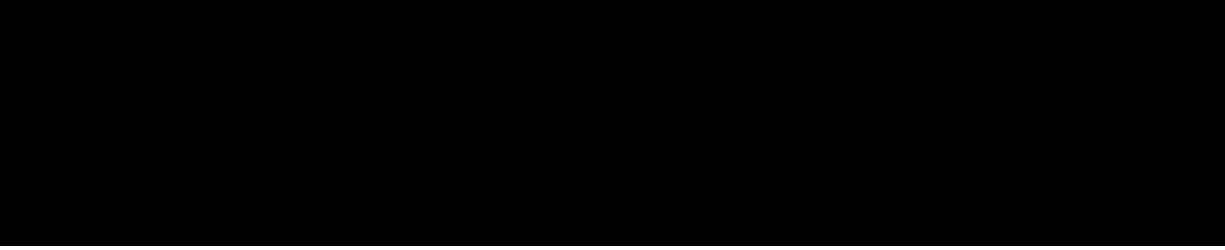
THE NATURAL COLORS IN PHOTOGRAPHY.

The subscriber begs to request those interested in the above discovery *not to visit him* until further notice, which will be given in the Daguerreian Journal. It is *absolutely essential* that he should have his time, and be left undisturbed. Stern necessity, arising from the fact that I can never complete my process if disturbed as hitherto, compels me to say both to friends and strangers, that my doors are locked, and will continue to be until I have completed my experiments. As this notice is given with a regard to the general good, it is presumed no offence will be taken.

L. L. HILL.

Westkill, Greene Co., N. Y.,
May 25, 1851.

Editors by inserting the above in their papers will confer a favor, and save many from spending their time and money to no purpose.



OUR DAGUERREOTYPES.

BUTLER, of this city, is now producing fine likenesses in oil. This gentlemen calls these pictures "Daguerreotypes in Oil." We never have seen better likenesses transferred from a Daguerreotype, and in an economical point of view, they are the best oil paintings that can be had.



E. LONG, of St. Louis, Mo. In our last, we spoke of H. H. Long as the gentleman who had produced likenesses of Jenny Lind. This was incorrect; we should have said E. Long. However, they both are a *long* ways from taking poor Daguerreotypes.



L. M. IVES, of Boston, will please pardon us (or rather our printer) for the mistake in the Journal in his name and locality. Mr. I. being himself a sure operator, will certainly pardon a failure in others. We will endeavor to do better in future.



N. E. SISSONS, of Albany. This gentleman has completed an extensive addition to his former establishment. We find here one of the most substantial proofs that close application and honorable dealing are awarded by success. Mr. S. has now five rooms—one for reception, a gallery or operating room, and three stock rooms. It is highly gratifying to his friends to learn of his success, and we predict for him a large and profitable business. We have ordered one of C. C. Harrison's best full sized camera tubes, and one of W. & W. H. Lewis' camera boxes, which will be forwarded to Mr. S., he being entitled to it from the fact that he has obtained for us the largest list of subscribers. He is a "practical operator."



J. D. WELLS, Northampton, Mass., has recently fitted up a large establishment in that place. Mr. W. is an old an experienced operator, and has five rooms in his establishment, a very fine sky and side light, and is prepared to execute such likenesses as will please the inhabitants of that beautiful village in the valley of the Connecticut.



SUBMARINE TELEGRAPH BETWEEN ENGLAND AND FRANCE.

The project of constructing a submarine telegraph between England and France, across the Straits of Dover, first announced during the year 1349, ^[A] has been in part accomplished. The following description of the laying down of the wire, we copy from an English Journal:—

^[A] See Annual of Scientific Discovery, 1850, page 128.

At one o'clock the steamer Goliath was ready to start across the Channel, with all the necessary apparatus on board, and a crew of about thirty men. Between the paddle-wheels, in the centre of the vessel, was a gigantic drum, or wheel, nearly fifteen feet long and seven feet in diameter, weighing seven tons, and fixed on a strong framework. Upon it was coiled up, in careful, close convulsions, about thirty miles of telegraphic wire, one-tenth of an inch in diameter, incased in a covering of gutta-percha, the thickness of the little finger. The point proposed to be reached, Cape Grinez, the nearest landmark to the English coast, and between Calais and Boulogne, is a distance of twenty-one miles, so that a surplus supply of nine miles of wire was held in reserve for the purpose of slackening. The connecting wires were placed in readiness at the Government pier in the harbor, and likewise at the Cape, where they run up the face of the acclivity, which is 194 feet above the sea-mark.

Some interesting experiments were first made upon a small scale to show the practicability of the plan. A mile of wire was paid out off the deck, from the pier to Shakspeare's Cliff, and the sinking process was proved to be a practicable performance. A communication was also sent through twenty four miles of wire. On Wednesday morning the experiment of sinking submarinely was practically commenced. The Goliath put out to the pier, with her telegraphic tackle and apparatus on board, under a calm sea and sky and a favoring wind. The connection between the thirty miles of telegraphic wire was then made good to 300 yards of the same wire inclosed in a leaden tube on shore, to prevent it being bruised by the shingle on the beach, and to enable the experimenters, as they proceeded out to sea, to send communications on shore. The vessel steamed out at the rate of three or four miles an hour into the open sea, in a direct track for Cape Grinez. The wire weighed five tons and the cylinder two. The operation of paying out the thirty miles of wire commenced on a signal to the sailors to "Go-ahead with the wheel, and pay out the wire," which was continuously streamed out over a roller at the stern of the vessel, the men at every 16th of a mile being busily engaged in riveting on to the wire,

square leaden clamps, or weights of iron, from 14 lbs. to 24 lbs. in weight, which had the effect of sinking the wire to the bottom, which, on the English coast commences at a depth of 30 feet, and goes on varying from that to 100 and 180 feet, which latter, or 30 fathoms, is the greatest depth.

The whole of the casting out and sinking was accomplished with great precision and success, owing to the favorable state of the day. The only conjectured difficulty on the route was at a point in midchannel, called the Ridge, between which and another inequality called the Varne, both well known and dreaded by navigators, there is a deep submarine valley, surrounded by shifting sands, the one being seventeen miles in length, and the other twelve, and in their vortex, not unlike the voracious one of Godwin Sands, ships encounter danger and lose their anchors, and trolling nets of fishermen are frequently lost. Over this, however, the wire was successfully submerged, below the reach, it is believed, of either ship's anchors, sea-animals, or fishing nets. The remainder of the route, though rougher on approaching the coast of France, was accomplished cleverly, but slowly. A communication, dated Cape Grinez, Coast of France, half past eight, P. M., and received at Dover by submarine telegraph, was as follows:—"The Goliath has just arrived in safety, and the complete connection of the under-water wire with that left at Dover this morning is being run up the face of the cliff; complimentary interchanges are passing between France and England, under the strait and through it, for the first time."

Notwithstanding this apparently successful result of the work, the line was cut asunder soon after the connection was completed on the rocks near Cape Grinez, the physical configuration of the French coast being very unfavorable. The precise point where the breakage took place was about two hundred yards out to sea, just where the twenty miles of electric line that had been laid down from Dover joins on to a leaden tube designed to protect it from the surge beating against the beach, and which also serves a similar purpose up the front of the cliff to the station upon the top. The leaden conductor, it would appear, was of too soft a texture to resist the oscillation of the sea, and thereby became detached from the coil of gutta-percha wire that was thought to have been safely encased in it. The occurrence was, of course, quickly detected by the sudden cessation of the series of communications, though it was at first a perplexing point to discover at what precise spot the wire was broken or at fault. This, however, was done by hauling up the line at intervals, a process which disclosed the gratifying fact, that, since its first sinking, it had remained in situ at the bottom of the sea, in consequence of the leaden weights or clamps that were strung to it at every sixteenth of a mile. The experiment, as far as it has gone, proves the possibility of the gutta percha wire resisting the action of the salt water, of the fact of its being a perfect waterproof insulator, and that the weights on the wire are sufficient to prevent it being drifted away by the currents, and for sinking it in the sands.

The work at present has been suspended, but will be resumed again during the spring of 1851; a somewhat different plan, however, has been proposed to be followed from that at first adopted. Instead of one slender wire, it is intended to lay down cables inclosing four lines. These cables will be composed of gutta-percha, four or five inches in thickness, the whole encased in wire rope, chemically prepared, to protect it from rot, and kyanized. There will be two of these cables, each twenty miles long, and three miles apart, the whole weight representing 400 tons; and it is expected, when chained down in the bottom of the sea, they will be of sufficient consistency and strength to resist the anchor of a 120-gun ship. The expense of the cables is estimated at £40,000. It is thought that the whole work may be accomplished by May, 1851.—*Annual of Scientific Discovery*.



ACTION OF SOLUTIONS OF CHLORIDES AND AIR ON MERCURY.

We have given in previous Numbers the results of M. Mialhe's experiments on the action of chlorides on some mercurial compounds, and he states that he had nearly concluded his experiments when it occurred to him to try whether mercury itself would not be acted upon by this class of substances.

Experiment, he states, confirmed his suspicions, for he found that the solutions of the alkaline chlorides put into contact with mercury and atmospheric air always produced bichloride of mercury, the quantity of which was greater in proportion to the concentration of the solution of the chloride, and the more perfect state of division of the metal, but no effect is produced unless oxygen, that of the air being sufficient, is present.

1st Experiment.—Mercury treated with the solution of alkaline chlorides (described in our last Number as the assay liquor), gave by stove heat 0·4 part of sublimate.

2nd Experiment.—The above repeated with the mercury finely divided by mucilage, yielded 0·7 part of sublimate.

The researches already detailed sufficiently prove, in the opinion of M. Mialhe, that the decomposing power of the alkaline chlorides is great, but they do not teach us anything as to their relative energy. The following experiments will supply this deficiency.

Hydrochlorate of Ammonia.—One hundred and twenty parts of hydrochlorate of ammonia and 30 parts of calomel were placed in an open bottle containing 1000 parts of distilled water, the temperature of which was gradually raised to 122° Fahr., and kept for half an hour; the sublimate produced amounted to 0·9 of a part.

The experiment repeated with the following salts gave the annexed quantities of sublimate:—

Chloride of Sodium,	0·4 of a part.
Chloride of Barium,	0·4 "
Chloride of Potassium,	0·3 "

It results from these experiments that the hydrochlorate of ammonia is the most powerful of these four salts.

In concluding his experiments, M. Mialhe remarks that the reactions which he has pointed out take place at common temperatures, but better at the temperature of the human body. All of them are produced in a short time, and some occur instantaneously, the greater part requiring only a few hours' contact for action. As then the different fluids contained in the human body contain oxygen, chloride of sodium, and hydrochlorate of ammonia, accompanied or not with hydrochloric and other acids which may facilitate their action, it follows that all the chemical phenomena produced under the circumstances described, occur in the human body when any mercurial preparation whatever is introduced into it; these always produce a certain quantity of corrosive sublimate in which their medicinal properties reside; and this fact explains, in the opinion of M. Mialhe, the hitherto unexplained physiological action and therapeutic properties of metallic mercury when introduced into the animal economy.—*Ann. de Chim. et de Phys.*, Juin 1842.



THE HEAT OF COMBINATIONS.

Every molecular change in the condition of matter is almost invariably connected with the evolution or absorption of heat, and the quantity of heat thus set free or absorbed bears always a definite relation to the amount of the mechanical or chemical action. To ascertain this relation has been the object of my investigations, and the following are a few of my principal results. 1. The solution of a salt in water is always accompanied by an absorption of heat. 2. If equal weights of the same salt be dissolved in succession in the same liquid, the heat absorbed will be less on each new addition of salt. 3. The heat absorbed by the solution of a salt in water holding other salts dissolved is generally less than that absorbed by its solution in water. 4. The heat absorbed by the solution of a salt in the dilute mineral acids is generally greater than that absorbed by its solution in water. In reference to the combination of acids and bases, the heat developed during the union is determined by the base, and not by the acid. An equivalent of the same base combined with different acids produces nearly the same quantity of heat. When a neutral salt is converted into an acid salt by combining with one or more equivalents of acids, no disengagement of heat occurs. When a double salt is formed by the union of two neutral salts, the same is the case, but when a neutral salt is converted into a basic salt, there is a disengagement of heat. When solutions of two neutral salts are mixed, and a precipitate formed from their mutual decomposition, there is always a disengagement of heat, which, though small, is perfectly definite in amount. The diamond disengages 7,824 units of heat during its combustion in oxygen gas, in the form of graphite, 7,778 units, and in that of wood charcoal, 8,080.—Dr. Andrews before the British Association at Birmingham.



[The following papers have been furnished us by Mr. Pirsson, one of the former editors of the *Eureka*, they having been previously published in that Journal; for the cuts, which will appear in our next number, we are indebted to Mr. Starr, one of the present editors of the *Farmer and Mechanic*, and formerly publisher of the *Eureka*.]



DAGUERREOTYPE.

BY JOHN JOHNSON.

As a general thing, however perfect any invention may be deemed by the inventor or discoverer, it falls to the lot of most, to be the subject of improvement and advancement, and especially is this the case with those new projects in science which open an untrodden field to the view of the artizan. Such has been in an eminent degree, the case with the discovery first announced to the world by Mons. Jean Jacques Claude Daguerre, of Paris, in the year 1839, and which excited unbounded astonishment, curiosity and surprise. It may be questioned had any other than Daguerre himself discovered a like beautiful combination, whether the world would have been favored with details exhibiting so much care, patience and perseverance as the Daguerreotype on its introduction. Shortly after these details reached the United States, by Professor S. F. B. Morse, of New York, who was, at the time of the discovery, residing in Paris. By this announcement, the whole scientific corps was set in operation, many repeating the experiments, following carefully the directions pointed out by Daguerre, as being necessary to success. Among the number in the United States, was Alexander S. Wolcott (since deceased) and myself, both of this city. On the morning of the 6th day of October, 1839, I took to A. Wolcott's residence, a full description of Daguerre's discovery, he being at the time engaged in the department of Mechanical Dentistry, on some work requiring his immediate attention, the work being promised at 2 P. M., that day; having, therefore, no opportunity to read the description for himself, (a thing he was accustomed to do at all times, when investigating any subject,) I read to him the paper, and proposed to him that if he would plan a camera, (a matter he was fully acquainted with, both theoretically and practically) I would obtain the materials as specified by Daguerre. This being agreed to, I departed for the purpose, and on my return to his shop, he handed me the sketch of a camera box, without at all explaining in what manner the lens was to be mounted. This I also undertook to procure. After 2 P. M., he had more leisure, when he proceeded to complete the camera, introducing for that purpose a reflector in the back of the box, and also to affix a plate holder on the inside, with a slide to obtain the focus of the plate, prepared after the manner of Daguerre. While Mr. Wolcott was engaged with the camera, I busied myself in polishing the silver plate, or rather silver plated copper, but ere reaching the end preparatory to iodizing, I found I had nearly or quite removed the silver surface from off the plate, and that

being the best piece of silver plated copper to be found, the first remedy at hand that suggested itself, was a burnisher, and a few strips were quickly burnished and polished. Meantime, the camera being finished, Mr. Wolcott, after reading for himself Daguerre's method of iodizing, prepared two plates, and placing them in the camera, guessed at the required time they should remain exposed to the action of the light; after mercurializing each in turn, and removing the iodized surface with a solution of common salt, two successful impressions were obtained, each unlike the other! Considerable surprise was excited by this result, for each plate was managed precisely like the other. On referring to Daguerre, no explanation was found for this strange result; time, however, revealed to us that one picture was positive, and the other negative. On this subject I shall have much to say during the progress of the work. Investigating the cause of this difference occupied the remainder of that day, However, another attempt was agreed upon, and the instruments, plates, &c., prepared and taken up into an attic room, in a position most favorable for light. Having duly arranged the camera, I sat for five minutes, and the result was a profile miniature, (a miniature in reality,) on a plate not quite three-eighths of an inch square. Thus, with much deliberation and study, passed the first day in Daguerreotype—little dreaming or knowing into what a labyrinth such a beginning was hastening us.

(To be continued.)



GALVANIZING DAGUERREOTYPE PLATES.

THE BATTERY. The best way is to purchase a good DANIEL'S BATTERY. Price \$2 00. To be had at any of the dealers.

For the benefit of those who cannot procure one, I would give the following directions for making one: Procure a copper cup, (sheet copper) 6 inches high, and 4 inches in diameter; also a cup of porous earthenware, (a small unglazed flower pot answers,) or a leather cup made water tight by sewing, or even a cylinder of porous wood, (say maple or ash,) 3 inches diameter, and about 6 inches high. This is to stand in the centre of the copper cup. Now, procure a solid cylinder of zinc, 6 inches high and 1 inch diameter. This is to stand in the centre of the porous cup. Fastened to the upper end of the zinc cylinder there should be a copper wire, pretty thick. It should be held by a screw, or soldered on, and previous to the fastening, both the wire and zinc should be made clean and bright, that there may be a perfect connection. A similar wire is to be fastened, in the same way, to the upper rim of the copper cup. Let these wires be about 6 inches long. To the end of the one coming from the copper fasten a plate of pure silver, about as large as a quarter plate, unless you wish to galvanize whole plates, in which case the silver plate must be larger, say, the size of a half plate. For small plates a silver dollar, enlarged to twice its size by hammering, will answer, but is not so good. To hammer it, it must be first heated, red hot, and allowed to cool slowly. This plate is to be immersed in the silver solution, described below, near the side of the solution jar. The other wire must be bent at the end like a hook, to receive the catch. This catch may be a piece of copper wire about 4 inches long, hooked at one end, and having several turns at the other end, to strengthen it. Between these turns the Daguerreotype plate, well polished, is inserted, as in a sort of jaw. The wire holding the catch should be so bent that the Daguerreotype plate will stand from one to three inches from, and face the anode, or silver plate.

Let the silver solution be well stirred, just before immersing a plate; then, blowing away the froth and scum, immediately dip the plate, and hang it on the wire. Let it remain until it takes on a deep blue color, take it out, grasp it with the plyers, rinse it freely with clean water, and dry it carefully with a spirit lamp. Buff again to a polish, galvanize to a light blue, rinse, dry, and buff again, and it is ready for the coating box.

Silver Solution. Dissolve in 1 quart of soft water, half a pound of *Cyanide of Potash*. In this dissolve the *Chloride of Silver* procurable from a silver dollar. Filter, through paper, or clean sponge, and it is ready for use, excepting that it will probably have to be reduced with water. It should be reduced till it works mellow, and free

from streaks. The occasional addition of a lump of cyanide will prevent a flowery deposit of oxide of silver. Occasionally, also, add a little chloride of silver, and more cyanide. The cyanide should always be in excess. The reason why this should be occasionally added, is that the solution becomes too strong, with the silver, from the anode. The connections must be kept bright, with a file or otherwise.

The manner of charging the above battery is as follows: Nearly fill the porous cup with water, and stir in about a tea-spoonful of sulphuric acid. Two or three drops of acid added once a week is enough. The copper cup should be filled with a saturated solution of sulphate of copper, (blue vitriol,) and the solution kept saturated by suspending in it a little sack of the blue vitriol.

☞ The zinc cylinder, previous to use, should be amalgamated, as follows: Place it in a plate, and brighten it by rubbing it with a swab, wet with dilute sulphuric acid. Then with the same swab, rub on mercury, until the whole surface is bright.

Chloride of Silver. Dissolve a silver dollar in about a gill of a mixture of nitric acid and water, equal parts, by the aid of a gentle heat. Let it cool. Throw it into an earthen or glass vessel, containing about 1 quart of strong salt water. Let the precipitate settle. Pour away the liquid, add a large quantity of water, let the chloride of silver settle, pour away the water, and repeat this at least fifty times. The residue is pure chloride of silver.

Any glass or earthen cup, of suitable shape and dimensions, will answer for a solution dish.

It should be remembered that a strong battery, and a strong solution require the plate to be kept at a greater distance from the anode. This distance will range from one to three inches.—*Hill's Treatise*.



ANSWERS TO CORRESPONDENTS.

NOTICE.—Those persons subscribing for this Journal will please bear in mind to write in a plain hand the name, town, county, and state. By observing the above you will save us much trouble, and, at the same time, receive the Journal with more promptness.

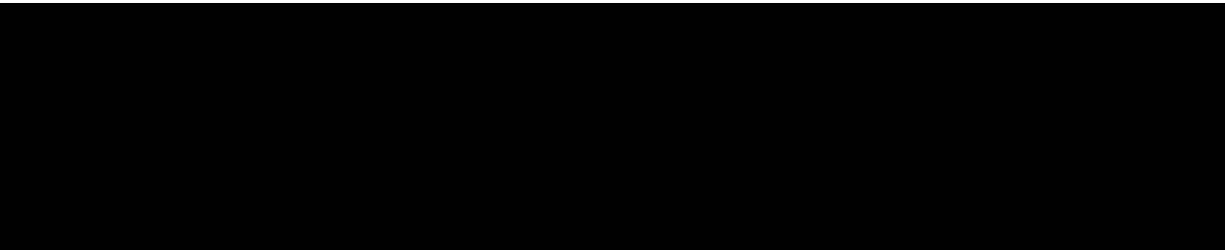
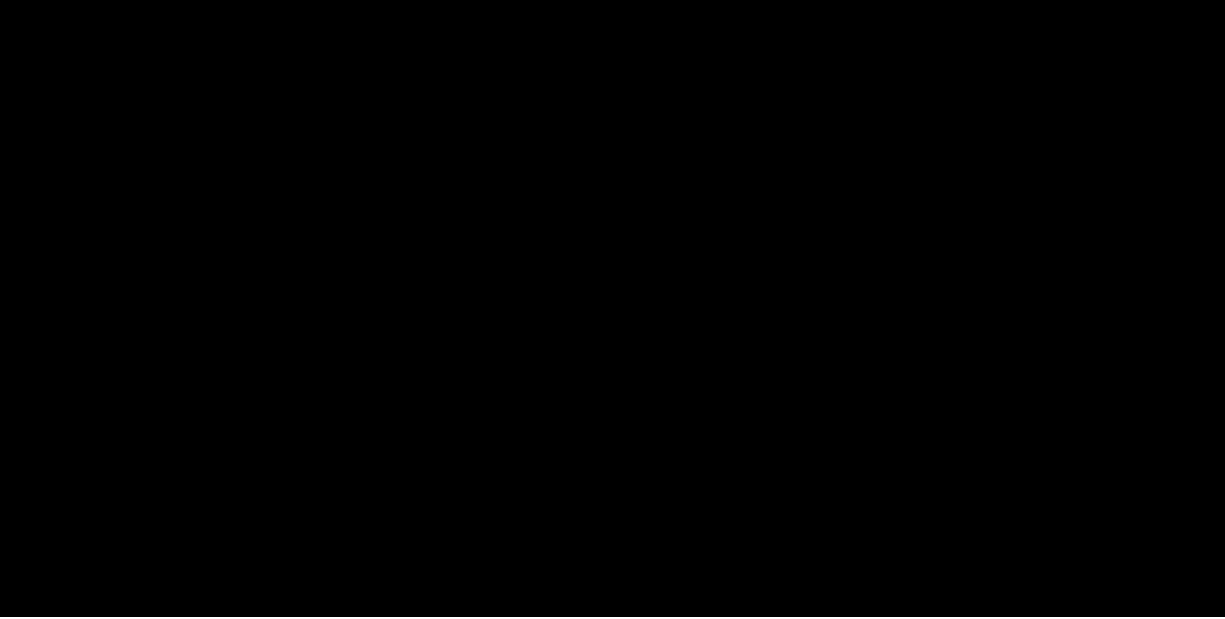
G. K. W., Mass.—We have been unable to prepare the chemicals you ordered. Our arrangements are somewhat extensive in the department referred to.

M. S. W., Ala.—Mr. W. says, "Will you inform me, through your Journal or otherwise, Is iodine entirely lost when it combines with a few drops of a solution of potassa!"

It is not. The following experiment will better illustrate:—Put a few grains of iodine into a clean and dry Florence flask, warm it gradually over a spirit lamp, and the iodine rises in a vapor of beautiful violet color. When cool, pour in "a few drops of a solution of potassa," the iodine disappears at once, and a solution of iodide of potassium is formed; this may be heated to redness without evolving any violet fumes, and yet it is certain that iodine is there, but not in a free but a combined state, and heat cannot expel it from combination with the potassium. Let the flask cool a little, and add a few drops of sulphuric acid; the violet fumes of iodine instantly appear; the sulphuric acid has induced the formation of sulphate of potassa, and expelled the iodine unchanged.

F. B. B., Ky.—Your money is received, and the first Vol. forwarded per order. Thank you for your hearty congratulations.

J. G. P. R. I.—Yours of the 26th in hand. We advise you for the present to say nothing of your discovery: it probably has more valuable importance than you are at present aware of.



ADVERTISEMENTS.

NOTICE TO ADVERTISERS.

As our Advertising List is already large, and the demand rapidly increasing, we are under the necessity of requesting those who wish to be made known through our Advertising Columns, to make their Advertisements as short as possible. The Daguerreian Journal is devoted to the interest of all.

DAGUERREIAN LIBRARY.

All of the works published on the Daguerreian Art, for sale at the Office of the Daguerreian Journal.

Address (*post-paid*) S. D. Humphrey,

No. 311 Broadway, New York.

DAGUERREOTYPE WORK,

PHOTOGRAPHIC RESEARCHES AND Manipulations, including the Author's former Treatises on Daguerreotypes,

By L. L. HILL.

The above work is now ready, price \$3 per copy.

Orders addressed to S. D. Humphrey,

No. 311 Broadway, N. Y.

WANTED.

A SITUATION by an experienced Operator. One who can come well recommended, and has been engaged in the Daguerreotype business for the last eight

years. Has no objection to any city in the United States.

For further information inquire at the Office of the Daguerreian Journal.

WANTED.

TO

DAGUERREOTYPISTS OF PHILADELPHIA.

WANTED—A situation, by an old and experienced Operator, to engage in some establishment in the above named city. The Advertiser can produce the best of Testimonials, and satisfactory specimens of Daguerreotypes.

For further information, inquire at the office of the Daguerreian Journal.

None need apply unless prepared to give a permanent situation to a good operator.

SCOVILL MANUFACTURING CO.,

DEALERS IN DAGUERREOTYPE MATERIALS,

WHOLESALE AND RETAIL,

No. 57 Maiden Lane, New York,

Agents for Voightlander & Sons, and C. C. Harrison's celebrated Cameras; Harrison's, Brinckerhoff's, and Lewis' Buffing-Wheels.

The Scovill Manufacturing Co. are now prepared to offer to the public, cases of their own manufacture, which, for style, workmanship and superior finish, are not excelled by any in the market, and it is their intention to put them at prices which cannot fail to satisfy both the dealer and operator.

They think it quite unnecessary to enumerate all the articles which they may have on hand for sale, but simply to state that they are prepared to furnish every description of Daguerreotype Goods of the best quality at low prices, and to fill all orders with promptness and dispatch.

May 15th, 1851.

MYRON SHEW,
DEALER IN DAGUERREOTYPE APPARATUS AND MATERIALS,
WHOLESALE AND RETAIL,

No. 118 Chestnut street, Philadelphia, Pa.

German and American Cameras of all sizes and of the best manufacture, Coating Boxes, Camera Stands, Mercury Baths, Buffing Vices, Head Rests, Plates, Cases, Chemicals, Frames, Gold and Gilt Locketts, Fancy Book Cases, Papier-Mache Cases, Preservers, Mats, Glass, and all articles used in the Art constantly on hand in great variety at the lowest prices.

Mr. S. having had a number of years practical experience in the Daguerreian Art, feels confident that he is prepared to give satisfaction to his customers in the selection of articles used in the business.

This being the only establishment in Philadelphia devoted exclusively to Daguerreotype materials. Artists will find it to their interest to call at once where they are sure of finding every article used in the Art.

Orders from the country promptly and faithfully attended to.

WANTED.—A Partner wanted in the Daguerreotype business. Any person who is a first-rate practical operator, and can command from five hundred to a thousand dollars, can now have one of the best opportunities of investing his capital in one of the best locations in a western city.

The establishment is now doing a very profitable business, and is capable of increasing to such an extent as to warrant the success necessary to induce a partner.

No one need apply unless he has the Cash Capital.

For further information inquire at the Office of the Daguerreian Journal.

CHEMICAL, LABORATORY.

DR. ANTISELL has Removed his Chemical Laboratory to 63 Franklin St., where he is prepared to undertake chemical analysis and examinations into the purity of materials.

Dr. A. will also take charge of any series of experiments to elucidate particular subjects.

Assays, analyses of ores, minerals, soils, and mineral water, made with correctness and despatch. May 1, 1851.

\$10 REWARD.

A LARGE SIZED (plate, 11 by 13 inches) Daguerreotype has been recently stolen from my door, supposed to have been taken by some person about visiting Europe. Said Daguerreotype was a large sized head, and is a likeness of myself.

Any person returning said picture, and the large gilt frame that contained it, shall receive the above reward.

J. GURNEY,

No. 189 Broadway, N. Y.

REMOVAL.

MYRON SHEW,

DEALER IN

DAGUERREOTYPE MATERIALS,

No. 118 Chestnut Street, Philadelphia.

MR. SHEW would respectfully inform his friends and customers that he has removed his place of business from his old stand. No. 116, to 118 Chestnut street, two doors from his former place, where he has made a permanent location, and, having greater facilities, will be better able to meet the increasing demands of his

business. No pains will be spared in the selection of articles used by Daguerreian Artists, and every effort used to give satisfaction to his customers. A good supply of all articles used in the business constantly on hand, comprising German and American Cameras, of all sizes, Camera Stands, Coating Boxes, Mercury Baths, Head Rests, Buffing Vices, Gilding Stands, Lamps, Plates, Cases, Frames, Gold and Gilt Locketts, a variety of Fancy Cases, &c., &c.

Mr. S. would take this opportunity of returning his sincere thanks for former patronage, and solicits a continuance of the same. All orders either from the city or country, promptly and faithfully attended to.

MYRON SHEW.

Philadelphia, June 1st, 1851.

DAGUERREOTYPE.

JOHN SAWYER,

Successor to WILLIAM SHEW, dealer in Daguerreotype Apparatus and Materials, wholesale and retail. Also, manufacturer of Cases.

No. 123 WASHINGTON STREET, BOSTON, MASS.

LOUIS BECKER

Well known CHEMICALS, for sale at BECKER & PIARDS, No. 201 Broadway, N. Y.

CLEVELAND DAGUERREOTYPE STOCK

DEPOT,

JOHNSON & FELLOWS,

SUPERIOR ST., CLEVELAND, O.,

Keep a large and well selected assortment of Daguerreotype Goods on hand, and for sale at the lowest New York prices.

We respectfully call the attention of operators to our assortment of goods, which we warrant genuine.

Every exertion will be made to keep the best, latest and cheapest goods on hand.

Our assortment consists of every variety of Cameras, Plates, Cases, Chemicals, Polishing Materials, Apparatus, Fixtures, Frames, Sockets, &c., &c.

Instructions in the Art are carefully given.

CHAS. E. JOHNSON,
PARKER FELLOWS.

NEW YORK

DAGUERREOTYPE

STOCK ESTABLISHMENT.

The undersigned would call the attention of Daguerreotypists and the trade to the recent improvements in their manufacture of plates, which, by a new method of finishing, enables them to present an article possessing a finer and softer surface, and requiring much less polishing than formerly. The edges of all are neatly bent.

Owing to greater facilities in manufacturing, the prices have been reduced, so as to bring them at about the same cost as the best imported brands.

They have constantly on hand a large assortment of

Plates,
Cases,
Frames,
Mattings,
Preservers,
Locketts,
Chemicals,

and Apparatus of every style and variety which they offer at the lowest prices.

Agents for Voightlander's, Harrison's and other Cameras; Harrison's new Buffing Wheel, &c., &c.

SCOVILL MANUFACTURING CO.,
No. 57 Maiden Lane, New York.

BENJAMIN FRENCH,

No. 109 Washington Street, Boston.

DAGUERREOTYPE Apparatus, Plates, Cases, Frames, Gold Locketts, Polishing materials, Chemicals, and every description of Goods used in the Daguerreotype business, constancy on hand and for sale, at wholesale and retail, at the lowest cash prices.

3tf

\$5 REWARD.

STOLEN from the door of Clark Brothers, 551 Broadway, one full size Daguerreotype View, in papier mache frame, oval fire gilt mat. Said View of a GOTHIC COTTAGE, on the steps of which can be seen a lady, two or three boys and a dog, Any person returning the above described Picture, or giving information where it may be found, shall receive the above reward Oct 16.

TWO NEW INVENTIONS

IN THE DAGUERREOTYPE ART.

"PECK'S PATENT PLATE HOLDER," and the "*Bent Edge Daguerreotype Plate*" used in connection with it. An instrument is sold for seventy-five cents, with which every operator can bend his own plates. The holder is a desideratum, and only requires to be used to be appreciated. It is so constructed that it will hold the plate through all the stages of cleaning, buffing, polishing, coating, taking the picture in the camera, and mercurializing without any change. During the whole process, the

plate need not be touched with the fingers, and does away with the use of wax, &c., &c.

The prices for the holders are mediums, \$1 00—quarters, \$1 50—Halves, \$2 00—whole size, \$2 50.

The "*Magic Back Ground*." The discovery of this is due Mr. C. J. ANTHONY, of Pittsburgh, Pa. By this process the most beautiful effects can be produced, and the pictures are set forth in bold relief on back grounds of various shapes and tints. Pictures taken with the "*Magic Back Ground*," will be emphatically the "*Pictures for the Million*." The Patent is applied for, and the right ratified upon the receipt of the Patent, for the sum of Twenty-Five Dollars.

L. CHAPMAN, 102 William St., N. Y.

Manufacturer of Cases, Mats, Preservers, Plates, Importer of the genuine Star Brand, No. 40 French Plates, and dealer in Daguerreotype stock generally.

CLARK BROTHERS, & B. L. HIGGINS.

Daguerreian Gallery, over the "Regulator," Franklin Buildings, Syracuse, N. Y., No. 128 Genesee St., Utica, Tremont Row Boston, and 551 Broadway, New York.

Likenesses by the improved Daguerreotype of various sizes, and of the most delicate executions may be obtained at the above rooms during the day, from 8 A M., to 5 P. M.

Chemicals, Plate, Cases, Cameras, Apparatus, and other materials, connected with the art, constantly on hand, and for sale at New York prices.

All articles are selected with great care and warranted to give the best satisfaction.

DAGUERREIAN ESTABLISHMENT.

JOHN ROACH, Optician, 79 Nassau Street, New York. Has always on hand

VOIGHTLANDER INSTRUMENTS, quarter, half, and whole size.

American Instruments, Roach's make, warranted to be superior to any yet made in the United States. They work with sharpness, and quickness, and persons purchasing can test them with the high priced German Instruments.

Coating Boxes, Flint Glass Jars, cemented in, and ground air tight.

Mercury Baths, with Thermometer Scale in front.

Head Rests, Stands, Cases, Chemicals, &c.

Plates, French 40th of the Star, and other first quality Brands.

Bromine Roach's Triple Compound, now called Quick-stuff, works with certainty and quickness, in all weather, and pictures taken with it have a rich white tone.

GALVANIC BATTERIES, &c.

CINCINNATI

DAGUERREOTYPE STOCK ESTABLISHMENT.

The undersigned would call the attention of those wishing Daguerreotype goods, to his large and general assortment of all articles appertaining to the Daguerreian arts. The following named articles comprise only a few, compared with the great variety of his stock.

Cameras—"Voightlander and Sohn's" German Cameras.

C. C. Harrison's Quarter, Halves, Wholes, Mammoth, View, and Quick Working Cameras! Also a general assortment from other manufacturers.

Complete sets of Apparatus—such as Coating Boxes, Mercury Baths, Plate Holders, Buffs, Buff Wheel, Peck's and Scovill's Patent Blocks, Lewis' new Iron Rests, Iron and Chair Head Rests, Gilding Stands.

Cases—Papier-Mache inlaid with Mother of Pearl, Velvet in and out with Clasps and Locks, Morocco lined with Velvet, Silk and Satin, Plain and Gilt, also a great variety of fancy patterns.

Plates of all sizes—Scovill's, French, 40th of the Star, and other first quality Brands.

Chemicals—American, German and French Bromine, Dry and Chloride of Iodine, Hydrofluoric Acid, Cyanide of Potassium, Pure Distilled Mercury, a superior article of Chloride of Gold for Gilding, Hyposulphite of Gold, Chloride of Calcium, Colors, Rotten Stone, Rouge, an excellent article.

Lamps, Nippers, Glass Funnel, Glass Graduates, Prepared Buckskins, Brushes, &c., &c.

All of the above articles are of the first quality, and will be furnished at reasonable rates.

Orders punctually attended to and satisfaction given.

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STOCK ESTABLISHMENT.

THE SUBSCRIBER, would respectfully inform the Daguerreian Artists, that he has constantly on hand a large assortment of Daguerreotype apparatus, plates, cases, and chemicals, comprising in part the following:

Voightlander & Sons, Harrison's, Lewis and Perry's Cameras and other apparatus of the most approved styles.

PLATES—Scovill's, Chapman's and the different brands of French plates.

CASES—Silk and velvet lined, Papier Mache and a great variety of fancy cases.

CHEMICALS—American, German and French Bromine, chloride of iodine, do gold, calcium, mercury, rouge, rotten stone, black polish, colours, brushes, rosewood and Papier mache, frames, mats, glass preservers, prepared buck skin, &c., &c. Every article used in the business, which I will furnish to operators at retail or wholesale, as low as the same quality of articles can be bought in New York or elsewhere.

Our motto is small profits and quick sales.

N. E. SISSON.

No. 496 Broadway, Albany N. Y.

LEVI CHAPMAN,

No. 102 William street, New-York, Manufacturer of, and Dealer in
Daguerreotype Stock.

Daguerreotype Cases.

Medium size, from \$15 to \$198 per gross,

Quarter " " 24 to 288 "

Half " " 60 to 432 "

MATS, PRESERVERS and CHEMICALS of all kinds. French and American
PLATES.

L. C. imports the genuine No. 40 Star Brand French Plate, which he keeps
constantly on hand, together with an assortment of other kinds.

PAPIER MACHE CASES, inlaid with Mother of Pearl, exceeding in beauty any
thing of the kind heretofore made, from 90 to \$1152 per gross.

L. C. is also proprietor of Peck's patent moveable Plate-holder.

GEORGE DABBS, }
JAMES CREMER, } Travelling Agents.

PREMIUM DAGUERREOTYPE DEPOT AND MANUFACTORY,

W. & W. H. LEWIS, 142 Chatham Street, New York, keep constantly on hand,
superior CAMERAS of all sizes; also, *quick working* Cameras, fully equal to any
imported. All kinds of Apparatus, including our Patent PLATE VICES and GILDING
STANDS; Galvanic Batteries for Electrotyping, for durability superior to all others.
Buffing Lathes, on the most approved plan, Plates, Cases, Chemicals, Polishing
Materials of every description. All kinds of Walnut, Rosewood and Gilt Frames for
Daguerreotypes, outside Show Frames, Diploma, Certificate and Picture Frames

made to order. Pressing Machines, for Straw and Leghorn Plats, of improved construction.

1tf

DAGUERREOTYPE FURBISHING ROOMS.

WM. A. WISONG,

No. 2 N. Liberty Street, Baltimore, Md.

HAS CONSTANTLY ON HAND, a full and general assortment of Stock, for Daguerreotype use.

Embracing every variety of Cameras, Plates, Cases, Fixtures, Trays, Chemicals, and Materials used by Daguerreian Artists, all of which are offered at the lowest market rates.

Orders from Artists are respectfully solicited and forwarded with dispatch.

2

Circular to Daguerreotypists.

GEORGE DABBS & JAMES CREMER, Travelling Agents for L. Chapman, 102 William street, New York, manufacturer of Daguerreotype cases, mats, preservers, and plates, and importer of the genuine Star brand, No. 40, French plate, and last, though of first importance, proprietor of "Peck's patent plate holder,"—the great desideratum which only requires to be used to be appreciated. Prices, \$1,00 for medium; \$1,50 for quarters; \$2,00 for half's and \$2,50 for whole size holders—a vice to hold the blocks \$1,50 and an instrument for bending the plates 75 cents. They would inform Daguerreotypists and dealers that they will wait upon as many during the winter, as their time will permit, for the purpose of exhibiting the patent Plate Holder, for a description of which see advertisement headed "Two New Inventions".

1tf

New York, November 1, 1850.

INSLEY'S DAGUERREOTYPE GALLERY

REMOVED TO

No. 311 BROADWAY, N. Y.

(Between Stewart's and the City Hospital.)

THIS, our new Sky-light Gallery, is located on the second floor at the above number, and is universally acknowledged to be the most convenient and effective Gallery in the City. Every real improvement is taken advantage of, and, aided by scientific and gentlemanly assistants, we trust our pictures cannot be excelled.

The Clergy—the Statesman—the Artist—the Man of science—and all lovers of really good Daguerreotypes, throughout the United States, are invited to call and examine our collection.

P. S.—This Gallery was for several years located on the corner of Cedar street, but is now removed to No. 311 Broadway.

CAMERAS.

C. C. HARRISON, Manufacturer of Cameras, and Camera Lenses, of all sizes and of the latest improvements. Office 293 Broadway, New York, where in his Gallery may be seen specimens of Daguerreotypes taken with instruments of his own manufacture, which for accuracy of performance are unsurpassed by any in the world.

N. B. A new and improved Camera for taking views, is manufactured by him, at a price unprecedentedly low.

C. C. HARRISON,

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2tf

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2000 LOUIS L. BISHOP'S superior PLATES, offered for sale at a great bargain,
by

VICTOR BISHOP, 23 Maiden Lane. N. B. These Plates are silvered by the
electro-magnetic process, and warranted superior to the best French plates.

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THE SUBSCRIBER still continues to carry on the business of ENGRAVING ON WOOD, in all its branches. His facilities are such that he is enabled to execute all orders promptly, and in every style of the Art, upon the most reasonable terms; while the experience of many years enables him to feel confidence in his efforts to give satisfaction to all who may favor him with their patronage.

Particular attention paid to the Drawing and Engraving of Subjects from Daguerreotypes.

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Few offices in the city have greater facilities for doing work with neatness and despatch, as most of the materials are new, and long editions are done by steam power presses.

☞ Daguerreian Journal Print.

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Broadbent, Samuel, 138 Chestnut st. Philadelphia.

Barnes, C., Mobile, Ala.

Bowman, Isreal, Berling, Canada.

Benson, W., Boonville, Mo.

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Bogardus, A., No. 217 Greenwich street, N. Y.

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Brown, H. S., Milwaukee, Wis.

Buxton, John, St. Catharine street, Montreal, Canada.

Bradlee, J. E., Boston, Mass.

Buell, E. M., Pittsfield, Mass.

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Bisbee, A., Dayton, Ohio.

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Geer & Benedict, Syracuse, N. Y.
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Hough & Anthony, Pittsburg, Alleghany Co., Pa.
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Williams, J. B., Philadelphia, Pa.

Williams, J. B., Philadelphia, Pa.

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THE DAGUERREIAN JOURNAL,

Devoted to the Daguerreian and Photogenic Arts,
also, embracing the Sciences, Arts and Literature.

The Daguerreian Journal is published semi-monthly, at 311 Broadway, on the 1st and 15th of every month.

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